

Bachelor's Thesis

Noisy Labels in Classification: A Bayesian Framework applied to Occupational Coding

Department of Statistics
Ludwig-Maximilians-Universität München

Julian Ehrenberg

Munich, January 27th, 2026



Submitted in fulfillment of the requirements for the degree:
“Bachelor of Science (B. Sc.)”

Supervised by Dr. Malte Schierholz

Abstract

In many applied classification settings, the observed labels are noisy measurements rather than error-free ground truths. Occupational coding is a prominent example: open-ended job descriptions are mapped to standardized codes by human coders and, increasingly, by automated tools. Yet, disagreement is common, and no external gold standard is available. This thesis develops and applies a Bayesian noisy measurement model for categorical ratings based on the Dawid and Skene latent class framework. The model jointly infers latent occupational classes, class prevalence, and source-specific confusion matrices, yielding posterior uncertainty at both the item and rater levels. Posterior inference is implemented in Stan, and Hamiltonian Monte Carlo diagnostics are used to assess convergence and stability.

Empirically, the model is applied to survey data coded by two professional raters and one automated tool for both classification standards, the *International Standard Classification of Occupations* (ISCO) and the German *Klassifikation der Berufe* (KldB). Focusing on the two category reductions, the analysis reveals patterns that are consistent with rater specific tendencies and non random tool errors, although conclusions remain conditional on the modeling assumptions and the chosen recoding. For ISCO (major groups 2 vs 3, Professionals vs Technicians and Associate Professionals), the tool's posterior confusion pattern is consistent with an over-assignment to class 2 which suggests weaker performance than both human raters in this dataset. For KldB (major groups 7 vs 8, Business organisation, accounting, law and administration vs Health, social services, teaching and education), the data imply very strong agreement, resulting in concentrated posterior class probabilities for most items. Several combinations of priors for class proportions and confusion matrices were evaluated, and the results show partial sensitivity, with noticeable quantitative shifts under stronger prevalence priors. Finally, posterior item probabilities provide a practical basis for triage, for example by prioritizing manual review for items with low maximum class probability.

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1 Introduction and Motivation

1.1 Problem context and motivation

Empirical applications frequently depend on categorical classifications generated by humans or algorithms, even though the underlying true category is unobserved. Typical examples include medical diagnoses, coding of open-ended survey responses, content moderation decisions, and relevance judgments in information retrieval. In such settings, recorded labels are often subject to error, disagreement, and ambiguity. They are therefore best viewed as noisy measurements of an unobserved latent state rather than as objective ground truth.

Probabilistic models of noisy labels offer an alternative perspective. Instead of treating one coder or a majority vote of coders as the gold standard, these models posit a latent true class for each item and explicit rater-specific error patterns. The classical work of Dawid and Skene (1979) introduced such a latent class model for categorical ratings. Related ideas were developed in medical diagnostics to infer disease prevalence and test characteristics without a gold standard (Joseph et al., 1995). While early applications focused mainly on medical contexts, more recent work has adapted and extended these models to crowdsourced annotation and natural language processing (Han et al., 2024). These approaches yield posterior distributions over the unobserved true labels and over parameters that describe rater quality. They therefore provide a principled way to quantify uncertainty and describe systematic differences between raters.

Occupational coding is a paradigmatic example of this problem. In large social surveys, respondents are typically asked to describe their occupation in one or two open-ended questions, which are later coded into standardized categories. This approach is commonly used in computer-assisted telephone interviews (CATI) as well as in web surveys (Schierholz et al., 2018). These descriptions are then mapped to standardized classification systems such as the *International Standard Classification of Occupations* (ISCO) (International Labour Office, 2012) or the German *Klassifikation der Berufe* (KldB) (Bundesagentur für Arbeit, 2010). The mapping is often performed by trained coders, and it can also be supported by automated tools that suggest codes or assign them directly. Even under favorable conditions, two human coders frequently disagree on the same case, and agreement between humans and automated systems can be lower (Chapter 3). Simple interrater agreement coefficients, such as Cohen’s kappa, summarize how often coders agree beyond what would be expected by chance, based solely on the cross-classification of their ratings. These measures treat all coders symmetrically and do not posit or estimate an explicit latent true code (Wirtz and Kutschmann, 2007). They therefore remain descriptive at the level of observed agreement and do not provide a probabilistic representation of the unobserved occupational category.

At the same time, occupational codes are not merely descriptive metadata. They are used as covariates and outcomes in substantive analyses, and they often serve as labels for downstream social and economic models (Schierholz et al., 2018). This frames occupational coding as a noisy measurement problem: the recorded code is an error-prone proxy for an unobserved occupational category. Ignoring uncertainty in the labels can therefore

bias parameter estimates, overstate the precision of substantive conclusions, and distort the evaluation of predictive models (Han et al., 2024). Moreover, misclassifications are structured rather than random due to human factors, as human coders tend to confuse specific groups of occupations instead of making uniformly random errors (Massing et al., 2019).

1.2 Objectives and research questions

This thesis takes up these ideas in the context of occupational coding. The primary aim is methodological: to develop and present a coherent Bayesian noisy measurement model for categorical ratings, building on the Dawid and Skene framework and related work in Bayesian latent class modeling. The thesis makes explicit how the model is specified, how its assumptions relate to common agreement measures, and how it can be implemented and diagnosed in a modern probabilistic programming environment. A second aim is to explore the usefulness of this framework in a realistic survey setting involving multiple human coders and an automated system. Specifically, the study illustrates what can be learned about latent occupational codes, rater-specific error patterns, and classification uncertainty, even when annotated data are limited. In this context, it also considers to what extent a Bayesian latent class model can learn rater quality parameters when the number of categories is large and the amount of observed data is limited, a situation that is typical in practice.

Substantively, the empirical application is based on a dataset of approximately 1,400 survey respondents, each coded by two human coders and one automated coding tool for both KldB and ISCO groupings (Simson et al., 2023). This setting can be interpreted as a plausible future workflow in which manual coding is complemented by automated tool output, with no external gold standard available. The model is fitted to this dataset to assess identifiability and convergence behavior in regimes with many categories and sparse observations.

Against this background, the thesis addresses the following research questions:

1. What is the structure of the Bayesian Noisy Measurement Model used in this thesis, and how does it extend the classical Dawid and Skene latent class model for occupational coding?
2. To what extent can rater-specific quality parameters be stably estimated in the present dataset using a Bayesian Dawid and Skene model when the available information is limited by small sample sizes and incomplete overlap of ratings? In particular, how robust are the posterior estimates of the rater-specific confusion matrices to plausible changes in the prior specification?
3. How informative and practically useful are the resulting probabilistic estimates of latent occupational codes and rater characteristics for occupational coding workflows? This includes their potential use for constructing probabilistic truth estimates, ranking records by uncertainty to target manual review or recoding, and monitoring the performance of human and automated coders over time.

Methodologically, the thesis aims to clarify the relationship between a classical latent class model, and modern Bayesian approaches implemented in probabilistic programming frameworks, specifically Stan (Stan Development Team, 2024). Empirically, it aims to provide a transparent case study of occupational coding that illustrates both the strengths and limitations of Bayesian noisy label models in a realistic high dimensional setting.

2 Methodological Framework

2.1 The classical Dawid and Skene model

The model by Dawid and Skene (1979) is a popular latent class model for categorical ratings in the absence of a gold standard. The basic setting is as follows.

Symbol	Description	Interpretation
$i = 1, \dots, I$	items	survey respondents' job descriptions
$k = 1, \dots, K$	labeling sources	human coders or an automated tool
$j = 1, \dots, J$	true class index	latent occupational code
$l = 1, \dots, J$	observed class index	reported code by source k

Both indices j and l range over $\{1, \dots, J\}$ because the observed label is assumed to take values in the same set of categories as the latent true class, that is, any observed code could in principle be the true code.

The model assumes that each item i has exactly one unobserved true class. In what follows, this assumption is used as a working assumption for occupational coding, since both KldB and ISCO are designed as highly specific classification grids with non-overlapping categories.

The latent true class can be represented by indicator variables

$$T_{ij} \in \{0, 1\}, \quad \sum_{j=1}^J T_{ij} = 1,$$

where $T_{ij} = 1$ means that item i truly belongs to class j . The distribution of the true classes in the population is given by

$$p_j = P(\text{true class is } j), \quad \sum_{j=1}^J p_j = 1. \quad (2.1)$$

Let $n_{il}^{(k)}$ denote the number of times observer k assigns category l to item i . In the simplest case with a single rating per observer and item, $n_{il}^{(k)}$ is a 0/1 indicator. This is the case in our given data setting. No rater has treated any person more than once. The key parameters of interest are the observer specific misclassification probabilities

$$\pi_{jl}^{(k)} = P(\text{observer } k \text{ records } l \mid \text{true class is } j), \quad (2.2)$$

for $j, l = 1, \dots, J$ and $k = 1, \dots, K$, with

$$\sum_{l=1}^J \pi_{jl}^{(k)} = 1 \quad \text{for every } j, k.$$

If $j = l$, then $\pi_{jj}^{(k)}$ is the probability that observer k correctly assigns class j . For each observer k the square matrix $\Pi^{(k)} = (\pi_{jl}^{(k)})_{j,l=1}^J$ can be interpreted as a confusion matrix

that describes individual error patterns: rows correspond to the true class j and columns to the recorded class l , and each row sums to one. The diagonal entries $\pi_{jj}^{(k)}$ represent correct classifications and are therefore expected to be larger than the off-diagonal entries, since the observers are assumed to provide informative ratings rather than random guesses. Off-diagonal entries $\pi_{jl}^{(k)}$ with $j \neq l$ capture how observer k tends to confuse class j with other categories.

The model relies on two strong main assumptions, which could both be violated in practice through various effects. These issues are discussed in detail in Section 2.4:

1. *Conditional independence and independent items.* Conditional on the latent true class, ratings are independent across labeling sources. Using the count indicators, this means that for $T_{ij} = 1$ the joint probability of the observed ratings factorizes across sources:

$$P\left(\{n_{il}^{(k)}\}_{k,l} \mid T_{ij} = 1\right) = \prod_{k=1}^K P\left(\{n_{il}^{(k)}\}_{l=1}^J \mid T_{ij} = 1\right).$$

In addition, items are assumed to be independent of each other. In the present application, each item is rated once per source.

2. *No item-by-source interaction.* There is no systematic interaction between a specific item and a specific labeling source beyond the latent true class. Equivalently, conditional on $T_{ij} = 1$, source k assigns category l with probability $\pi_{jl}^{(k)}$:

$$P\left(n_{il}^{(k)} = 1 \mid T_{ij} = 1\right) = \pi_{jl}^{(k)}.$$

Thus, misclassification probabilities depend on the true class j and the source k , but not on the particular item i .

Together, they imply that, given the true class of item i is j , the joint distribution of all ratings for this item factorizes as

$$P\left(\{n_{il}^{(k)}\}_{k,l} \mid T_{ij} = 1\right) \propto \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}}. \quad (2.3)$$

This product form reflects that, conditional on the true class, each of the $n_{il}^{(k)}$ independent ratings that observer k assigns to category l contributes a factor $\pi_{jl}^{(k)}$ to the likelihood, so $\pi_{jl}^{(k)}$ appears raised to the power $n_{il}^{(k)}$. Since in the present case, with at most one rating per observer and item, $n_{il}^{(k)} \in \{0, 1\}$. Thus, $(\pi_{jl}^{(k)})^{n_{il}^{(k)}}$ equals 1 if no rating is given and equals $\pi_{jl}^{(k)}$ if a single rating in category l is observed. Note the proportionality sign, which indicates that multinomial coefficients (which depend only on the observed counts $n_{il}^{(k)}$ and not on the parameters p or π) are omitted because they do not affect maximum likelihood estimation (Kauermann, 2023).

Formally, the model is identifiable only up to permutations of the class labels: the likelihood is unchanged under any relabeling of the latent classes, implying $J!$ equivalent

maximizers corresponding to the same solution with different label orderings (Dawid and Skene, 1979). In applied work, this label switching ambiguity is typically handled by adopting a sensible labeling convention; for example, by requiring diagonal dominance of the confusion matrices, i.e., $\pi_{jj}^{(k)} > \pi_{jl}^{(k)}$ for $j \neq l$ for most j and k (Dawid and Skene, 1979).

To estimate p_j and $\pi_{jl}^{(k)}$, the likelihood of the observed ratings under the model is considered. Here, the likelihood is the joint probability of the observed ratings $\{n_{il}^{(k)}\}$ viewed as a function of the unknown parameters (p, π) , with the data treated as fixed. This likelihood is simplest to express in complete-data form if the true class indicators T_{ij} were observed. In practice, however, the true classes are latent and can be treated as missing data in this likelihood framework (Dawid and Skene, 1979).

If the complete data $(T_{ij}, n_{il}^{(k)})$ were available, the joint likelihood of all items would factorize as

$$L_c(p, \pi; T, n) \propto \prod_{i=1}^I \prod_{j=1}^J \left\{ p_j \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}} \right\}^{T_{ij}}. \quad (2.4)$$

This expression is obtained by extending Equation (2.3) from a single item with a fixed true class j to all items and all possible classes. By the conditional independence assumptions, the likelihood factorizes over items and observers. Each class-specific term is weighted by its prevalence p_j , and the indicator T_{ij} selects the unique true class of item i .

Taking logarithms (Kauermann, 2023) yields the complete data log likelihood:

$$\begin{aligned} \ell_c(p, \pi; T, n) &= \log L_c(p, \pi; T, n) \\ &= \log \left[\prod_{i=1}^I \prod_{j=1}^J \left\{ p_j \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}} \right\}^{T_{ij}} \right] + \text{const.} \\ &= \sum_{i=1}^I \sum_{j=1}^J T_{ij} \log \left(p_j \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}} \right) + \text{const.} \\ &= \sum_{i=1}^I \sum_{j=1}^J T_{ij} \left[\log p_j + \sum_{k=1}^K \sum_{l=1}^J n_{il}^{(k)} \log \pi_{jl}^{(k)} \right] + \text{const.} \\ \ell_c(p, \pi; T, n) &= \sum_{i=1}^I \sum_{j=1}^J T_{ij} \left[\log p_j + \sum_{k=1}^K \sum_{l=1}^J n_{il}^{(k)} \log \pi_{jl}^{(k)} \right] + \text{const.} \end{aligned} \quad (2.5)$$

Maximizing ℓ_c subject to the constraints $\sum_{j=1}^J p_j = 1$ and $\sum_{l=1}^J \pi_{jl}^{(k)} = 1$ for all j and k (which ensure that p and each row of $\Pi^{(k)}$ form valid probability distributions) leads to simple relative frequency estimators for p_j and $\pi_{jl}^{(k)}$ (Dawid and Skene, 1979). One convenient way to obtain the closed-form updates is via a Lagrange-multiplier argument (e.g. Bishop (2006), Appendix E). For reference, the derivation is included in Appendix A.1.

The resulting estimators are

$$\hat{p}_j = \frac{1}{I} \sum_{i=1}^I T_{ij}, \quad \hat{\pi}_{jl}^{(k)} = \frac{\sum_{i=1}^I T_{ij} n_{il}^{(k)}}{\sum_{l'=1}^J \sum_{i=1}^I T_{ij} n_{il'}^{(k)}}. \quad (2.6)$$

Observed data likelihood

In practice, the true class indicators T_{ij} are unobserved (latent), and inference has to be based on the observed ratings $n_{il}^{(k)}$ alone. The observed data likelihood is obtained by summing the complete data likelihood over all possible values of T_{ij} that satisfy $\sum_{j=1}^J T_{ij} = 1$ for each item i ,

$$L(p, \pi; n) \propto \prod_{i=1}^I \sum_{j=1}^J p_j \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}}. \quad (2.7)$$

Intuitively, because the true class is unknown for each item, its contribution to the likelihood is obtained by averaging the class-conditional likelihood over all J possible classes, weighted by their prevalences p_j . A multinomial distribution describes category counts given category probabilities. The kernel of a likelihood is the factor that contains all dependence on the parameters, dropping multiplicative constants that depend only on the data. Here, the product $\prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}}$ is the kernel of a multinomial likelihood for the categorical counts $n_{il}^{(k)}$ with category probabilities $(\pi_{j1}^{(k)}, \dots, \pi_{jJ}^{(k)})$, see for example Bishop (2006, Section 2.2). In this incomplete data setting, the maximum likelihood estimators no longer have a closed form.

EM algorithm

Following the general EM framework of Dempster et al. (1977), Dawid and Skene propose maximizing the observed-data likelihood $L(p, \pi; n)$ in (2.7) by treating the latent indicators T_{ij} as missing data. While T_{ij} is marginalized out and thus absent from the observed-data likelihood expression in (2.7), the EM algorithm makes the maximization tractable by iteratively working with the expected complete-data log-likelihood (2.5). The algorithm alternates between an E step and an M step.

The E step computes the posterior probability (the expected value) that item i belongs to true class j , given the observed ratings and the current parameter estimates p and π . By applying Bayes' theorem, the weights w_{ij} are calculated as:

$$w_{ij} = \mathbb{E}[T_{ij} \mid \{n_{il}^{(k)}\}_{k,l}, p, \pi] = \frac{p_j \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}}}{\sum_{q=1}^J p_q \prod_{k=1}^K \prod_{l=1}^J (\pi_{ql}^{(k)})^{n_{il}^{(k)}}}, \quad (2.8)$$

where the numerator represents the joint probability of the latent class j and the observed labels, and the denominator provides the necessary normalization across all possible classes $q = 1, \dots, J$ (Dawid and Skene, 1979). These weights represent the current best estimate of the unobserved true classes.

In the M step, the expected complete-data log-likelihood is maximized with respect to p and π . The resulting update equations take the same form as the complete-data estimators

in (2.6), with the unobserved indicator variables T_{ij} replaced by the posterior class weights w_{ij} calculated in the preceding E step:

$$p_j^{new} = \frac{1}{I} \sum_{i=1}^I w_{ij}, \quad (2.9)$$

$$\pi_{jl}^{(k),new} = \frac{\sum_{i=1}^I w_{ij} n_{il}^{(k)}}{\sum_{i=1}^I w_{ij} n_i^{(k)}}, \quad (2.10)$$

where $n_{i\cdot}^{(k)} = \sum_{l=1}^J n_{il}^{(k)}$ is the total number of ratings that observer k provides for item i (Dawid and Skene, 1979).

Repeating the E and M steps increases the observed-data log likelihood in each iteration and converges to a local optimum (Dawid and Skene, 1979). Because of symmetries in the likelihood, the solution can depend on initialization; Dawid and Skene therefore recommend data-based starting values and multiple restarts. In the Bayesian formulation in Section 2.2, inference does not rely on these EM updates. In Stan, the discrete latent class indicators are marginalized out and the continuous parameters are inferred from the posterior using Hamiltonian Monte Carlo.

Overall, the point estimates obtained via the EM algorithm serve as a frequentist benchmark for the Bayesian formulation introduced in Section 2.2. Both approaches share the same likelihood structure. However, the high dimensionality of the Dawid and Skene model, comprising $(J-1)(JK+1)$ free parameters, poses significant challenges for maximum likelihood estimation. In realistic sample sizes, this can lead to large and strongly correlated standard errors (Dawid and Skene, 1979), making the estimates sensitive to individual observations and necessitating manual robustness checks. These limitations underscore that quantifying uncertainty and sensitivity is central in practice. The Bayesian framework addressed in the following sections makes this uncertainty explicit through the posterior distribution and can incorporate prior distributions to provide necessary regularisation in sparse data settings.

2.2 Bayesian formulation

The Bayesian noisy measurement model used in this thesis keeps the likelihood structure of the Dawid and Skene model from Section 2.1 but treats the unknown quantities as random variables with prior distributions. In particular, the class proportions $p = (p_1, \dots, p_J)$, the rater specific confusion matrices $\Pi^{(k)} = (\pi_{jl}^{(k)})_{j,l=1}^J$, and the latent true classes T_{ij} are given a joint prior. Bayesian latent class models of this form have been widely used in diagnostic testing and annotation problems, for example to estimate disease prevalence and test characteristics without a gold standard (Joseph et al., 1995), to analyse crowdsourced labels and rater accuracies (Paun et al., 2018, Lakkaraju et al., 2015), and to extend the Dawid and Skene likelihood with item difficulty (Han et al., 2024).

A convenient way to describe the model is in terms of a generative process. This process follows a hierarchical structure: With the Dirichlet priors, probability parameters are generated, which, in turn, govern the Categorical realization of the latent class and the

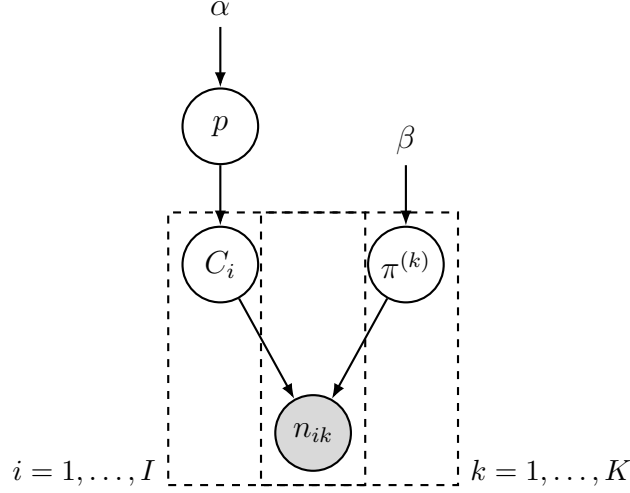


Figure 1: Graphical model (plate notation) of the Bayesian Noisy Measurement Model. p represents the vector of class proportions, C_i denotes the latent true class of item i , $\pi^{(k)}$ the confusion matrix of rater k , and n_{ik} the observed ratings. The hyperparameters α and β govern the priors.

Multinomial realization of the observed ratings. Let $C_i \in \{1, \dots, J\}$ denote the latent true class of item i and let $T_{ij} = \mathbb{I}(C_i = j)$ be the corresponding indicator representation, where $\mathbb{I}(\cdot)$ is the indicator function. The Bayesian formulation assumes:

1. *Class proportions.* The vector of class proportions p follows a Dirichlet prior

$$p \sim \text{Dirichlet}(\alpha),$$

where $\alpha = (\alpha_1, \dots, \alpha_J)$ is a vector of positive hyperparameters. The Dirichlet distribution serves as a distribution over probability vectors, ensuring that the elements of p sum to one. A common neutral choice is a symmetric Dirichlet prior, $\alpha_j = \alpha_0$ for all j , which treats classes exchangeably and does not encode a priori preferences for any particular class. Similar choices are used, for example, by Paun et al. (2018); in the binary case, Joseph et al. (1995) use a uniform prior on prevalence, i.e., $\text{Beta}(1, 1)$, which corresponds to $\text{Dirichlet}(1, 1)$.

2. *Latent true classes.* Conditional on p , the latent true class of each item is drawn independently from a categorical distribution,

$$C_i \mid p \sim \text{Categorical}(p), \quad i = 1, \dots, I.$$

The Categorical distribution represents the outcome of a single trial with J possible categories (equivalent to a Multinomial distribution with size $n = 1$). It models the single realization of the item's true class based on the population proportions p .

3. *Rater specific confusion matrices.* For each observer k and each true class j , the row vector of misclassification probabilities $\pi_{j\cdot}^{(k)} = (\pi_{j1}^{(k)}, \dots, \pi_{jJ}^{(k)})$ is given a Dirichlet prior

$$\pi_{j\cdot}^{(k)} \sim \text{Dirichlet}(\beta_{j\cdot}^{(k)}),$$

where $\beta_j^{(k)} = (\beta_{j1}^{(k)}, \dots, \beta_{jJ}^{(k)})$ is a vector of positive hyperparameters. Similar to the class proportions, this prior generates valid probability vectors for the rows of the confusion matrix. To encode the belief that observers are more likely to be correct than to make arbitrary mistakes, the diagonal element $\beta_{jj}^{(k)}$ is chosen larger than the off diagonal elements $\beta_{jl}^{(k)}$ for $l \neq j$, which yields a prior that favours confusion matrices with relatively large diagonal entries. In the empirical analysis all observers are given the same hyperparameters, so that the prior is exchangeable across raters.

4. *Observed ratings.* Conditional on the true class $C_i = j$ and on the confusion matrices $\Pi^{(k)}$, the observed rating counts follow the same multinomial sampling model as in Section 2.1. For each item i and observer k ,

$$(n_{i1}^{(k)}, \dots, n_{iJ}^{(k)}) \mid C_i = j, \pi_j^{(k)} \sim \text{Multinomial}(n_{i\cdot}^{(k)}, \pi_j^{(k)}),$$

where $n_{i\cdot}^{(k)} = \sum_{l=1}^J n_{il}^{(k)}$ is the total number of ratings that observer k gives to item i . This distribution models the counts of labels generated by the rater as the outcome of $n_{i\cdot}^{(k)}$ independent trials, governed by the specific error probabilities defined in $\pi_j^{(k)}$.

Combining these components yields the complete-data joint distribution

$$p(C, n, p, \pi) = p(p) p(\pi) \prod_{i=1}^I p(C_i \mid p) \prod_{i=1}^I \prod_{k=1}^K p(\{n_{il}^{(k)}\}_{l=1}^J \mid C_i, \Pi^{(k)}). \quad (2.11)$$

Integrating out (or marginalizing) the latent classes $C = (C_1, \dots, C_I)$ means that the joint distribution is summed over all J possible categories for each item i . Since the latent classes are discrete, this “integration” is mathematically a summation: the total probability of the observed ratings is obtained by summing the likelihood contributions across all possible latent categories, weighted by their prior probability p_j . This marginalization step eliminates the discrete parameters from the joint density and yields the observed-data likelihood $L(p, \pi; n)$ from Section 2.1. The marginalized posterior distribution for the continuous parameters is therefore

$$p(p, \pi \mid n) \propto L(p, \pi; n) p(p) p(\pi). \quad (2.12)$$

Posterior class probabilities for the latent labels can be reconstructed as derived quantities via

$$P(C_i = j \mid n, p, \pi) \propto p_j \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}}, \quad j = 1, \dots, J, \quad (2.13)$$

with $T_{ij} = \mathbb{I}(C_i = j)$ providing the corresponding indicator representation.

This posterior combines the information from the observed ratings with the prior assumptions about class proportions and rater error structures. The resulting Bayesian latent class model is closely related to diagnostic models without a gold standard (Joseph et al., 1995) and to Bayesian extensions of the Dawid and Skene framework used in annotation studies (Paun et al., 2018).

2.3 Posterior inference and Stan implementation

Exact analytic expressions for the marginalized posterior distribution $p(p, \pi \mid n)$ are not available. Instead, inference proceeds by Markov Chain Monte Carlo sampling from the continuous parameters p and π and by reconstructing posterior class probabilities for the latent classes T_{ij} as a derived quantity. In this thesis, Stan version 2.36 (Stan Development Team, 2024) is used for the implementation of the model. Stan is widely used statistical software, optimized for implementing Bayesian statistical methods. Moreover, Bob Carpenter, a main contributor to Stan, has co-published highly relevant academic papers within the past few years on Bayesian models of categorical data (e.g. Paun et al. (2018)). The key step that makes this feasible in Stan is to marginalize out the discrete latent variables C_i (or equivalently T_{ij}) from the joint posterior, so that the sampler only operates on continuous parameters (Stan Development Team, 2024). In this context, marginalization means explicitly summing the joint probability over all J possible values of the latent class C_i . By applying the law of total probability, the likelihood contribution of each item is transformed into a weighted sum (a finite mixture) of the conditional likelihoods for each possible class. This operation effectively eliminates the discrete parameters from the sampling process, leaving a likelihood function that is fully differentiable with respect to the continuous parameters p and π .

Using the multinomial model from Section 2.1, the marginal likelihood of the observed ratings is

$$L(p, \pi; n) = \prod_{i=1}^I \sum_{j=1}^J p_j \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}}, \quad (2.14)$$

which is exactly the observed data likelihood derived from the frequentist Dawid and Skene model (Dawid and Skene, 1979). Combining this likelihood with the Dirichlet priors for p and π yields a posterior density $p(p, \pi \mid n)$ over the simplex constrained parameters, i.e., probability vectors with nonnegative entries that sum to one. In Stan, the vectors p and $\pi_j^{(k)}$ are declared as simplex parameters, which enforces the constraints $\sum_{j=1}^J p_j = 1$ and $\sum_{l=1}^J \pi_{jl}^{(k)} = 1$ directly in the parameterization (Stan Development Team, 2024).

Posterior sampling is performed with Hamiltonian Monte Carlo, specifically the No U Turn Sampler as implemented in Stan (Stan Development Team, 2024). Multiple independent chains are run from dispersed initial values. Standard convergence diagnostics are monitored, in particular the potential scale reduction factor $\hat{R}$ close to one, sufficient effective sample sizes for all parameters, and the absence of divergent transitions or extreme values of the energy based effective sample size. These diagnostics are reported and discussed in Section 4.3.

Given the posterior draws of p and π , the posterior distribution of the latent class indicators T_{ij} is obtained in a generated quantities block by applying Bayes theorem for each item and class,

$$w_{ij} = P(T_{ij} = 1 \mid \{n_{il}^{(k)}\}_{k,l}, p, \pi) = \frac{p_j \prod_{k=1}^K \prod_{l=1}^J (\pi_{jl}^{(k)})^{n_{il}^{(k)}}}{\sum_{h=1}^J p_h \prod_{k=1}^K \prod_{l=1}^J (\pi_{hl}^{(k)})^{n_{il}^{(k)}}}, \quad (2.15)$$

which is the same expression as in the E step of the EM algorithm in Section 2.1 (Dempster et al., 1977, Dawid and Skene, 1979). The quantities w_{ij} represent posterior class probabilities for each item and are the basis for probabilistic truth estimates, uncertainty measures, and summaries of latent occupational codes in the application.

Compared to classical EM estimation (Dempster et al., 1977, Dawid and Skene, 1979), the Bayesian MCMC approach has the advantage that it provides full posterior distributions for class proportions and confusion matrices rather than single point estimates. This is particularly important in settings with many categories and limited data, where parameter uncertainty can be substantial and where regularizing priors can stabilize estimation (Joseph et al., 1995, Paun et al., 2018, Han et al., 2024).

2.4 Model assumptions and limitations

The Bayesian noisy measurement model inherits the structural assumptions of the Dawid and Skene likelihood, see Section 2.1. These assumptions are strong and should not be treated as empirically guaranteed in occupational coding settings.

First, the model relies on the assumption that each item belongs to exactly one latent true class. While classification systems such as ISCO or KldB are designed to provide mutually exclusive categories without overlap, this assumption is frequently challenged in practice. Real-world job descriptions are often vague, comprise multiple distinct tasks, or straddle the boundaries between closely related codes. In such cases, a single-class representation serves at best as an approximation. Consequently, posterior uncertainty should be interpreted as a signal of intrinsic ambiguity rather than solely as rater error.

Second, conditional on the true class, all ratings are assumed to be independent across observers and there is no item by observer interaction. Formally, the likelihood assumes

$$P\left(y_i^{(1)}, \dots, y_i^{(K)} \mid C_i = j\right) = \prod_{k=1}^K P\left(y_i^{(k)} \mid C_i = j\right),$$

and each rater is characterized by a confusion matrix with entries

$$\pi_{j\ell}^{(k)} = P\left(y_i^{(k)} = \ell \mid C_i = j\right), \quad \sum_{\ell=1}^J \pi_{j\ell}^{(k)} = 1 \quad \text{for all } j, k.$$

This implies that systematic disagreement is entirely attributed to rater-specific error patterns encoded in $\Pi^{(k)}$, while residual dependence between raters and item-specific difficulty are not modeled. In practice, this conditional independence assumption can fail. For example, professional coding institutes may exhibit shared conventions and decision rules (house effects), which induce correlated errors within an institute (Massing et al., 2019). In addition, procedural features can create dependence across items, for instance when cases are processed in ordered batches (e.g., alphabetically sorted job descriptions), which may influence rater attention or calibration over time. Such mechanisms would manifest as dependence and item-level heterogeneity that the baseline Dawid and Skene structure cannot represent.

Third, rater behavior is assumed to be stable across the sample, so the misclassification probabilities do not depend on time, survey wave, or item covariates. Any systematic variation of rater performance across subsets of items is therefore absorbed into the average confusion matrices and can bias the interpretation of $\Pi^{(k)}$ as a fixed rater quality descriptor.

From a Bayesian perspective, these assumptions are encoded in the likelihood, while the priors provide regularization for p and $\Pi^{(k)}$. Dirichlet priors are placed on the rows of $\Pi^{(k)}$ and can be set to concentrate mass on the diagonal, reflecting the belief that raters assign the correct class more often than an incorrect one (Joseph et al., 1995). This regularization reduces the plausibility of near-random labeling patterns a priori while still allowing asymmetric error structures if supported by the data.

Several limitations follow. The model is only identifiable up to permutations of the class labels, and the parameter space grows quickly with the number of classes and raters. As in the frequentist Dawid and Skene model, the number of free parameters scales with $(J - 1)(JK + 1)$, which can lead to substantial posterior uncertainty and strong dependencies between parameters when the number of multiply coded items is limited (Paun et al., 2018).

Most importantly, violations of conditional independence or unmodeled item difficulty can make seemingly stable posterior summaries misleading. Extensions that relax local independence, introduce item difficulty, or incorporate covariates are available and discussed conceptually in Section 6, but they are not implemented in the core model to keep the parameterization tractable and to focus on the baseline Bayesian noisy-label structure. Results in Sections 4 and 5 should therefore be interpreted as conditional on the Dawid and Skene assumptions, rather than as assumption-free measures of rater accuracy.

3 Occupational coding: Data and Disagreement

3.1 Occupational Coding and sources of disagreement

Occupational coding is the process of translating respondents' self reported job titles and task descriptions, typically collected via open ended survey questions, into standardized occupational categories. This step is necessary for statistical analysis and comparability across studies, but it is also a frequent source of measurement error because the mapping from free text answers to a discrete code is not always unique and often depends on contextual information that may be missing or vague. Schierholz et al. (2018) describe occupational coding in survey settings as costly and error prone and discuss how disagreement arises when coding is performed on the basis of open ended answers.

Code structures: KldB 2010 and ISCO-08. In the German context, occupations are often coded using the *Klassifikation der Berufe 2010* (KldB 2010) provided by the Federal Employment Agency (Bundesagentur für Arbeit), which assigns a five digit hierarchical code. The digits correspond to successive levels of aggregation: occupational area (1st digit), occupational main group (2nd), occupational group (3rd), occupational sub group (4th), and occupational type (5th) (Bundesagentur für Arbeit, 2010). For international comparability, occupational information is frequently coded or crosswalked to the *International Standard Classification of Occupations 2008* (ISCO-08), which is organized as a four level hierarchy with 1 digit major groups, 2 digit sub major groups, 3 digit minor groups, and 4 digit unit groups (International Labour Office, 2012). Conceptually, the KldB 2010 primarily groups occupations according to areas of activity and then differentiates according to skill level, while ISCO-08 is more explicitly structured according to skill level and skill specialization. It is important to note this difference. Exceptions hold, for example the first digit "0" represents Military activity in both classification systems. A detailed list on the meaning of each digit can be found in A.2.

Why coders disagree. Disagreement in occupational coding often arises because responses to open-ended occupation questions can be vague, ambiguous, or even internally contradictory, making the assignment of a specific code a complex matter of interpretation (Massing et al., 2019). Elias (1997) emphasizes that occupational classification is an inherently difficult and inexact process, typically characterized by relatively low levels of reliability. According to this view, misinterpretations often stem from a misunderstanding of the conceptual basis of classification systems like ISCO-88 (a predecessor of ISCO-08), or from national contexts where occupations are grouped by criteria other than the standard dimensions of skill level and specialization. In addition, coding outcomes depend on the specific rules and procedures applied, including agency-specific conventions and decision rules (so called house effects). Manuals cannot fully specify how to resolve all problematic distinctions, which leads to boundary problems between groups (Massing et al., 2019, Elias, 1997). Different agencies may also differ in whether and how they use auxiliary information and in how they resolve ties between multiple plausible codes. These mechanisms motivate modeling occupational codes as noisy labels that arise from rater-specific error patterns rather than treating observed codes as error-free.

3.2 Data Setting and preparation

The empirical data analyzed in this thesis are derived from an unpublished usability study conducted by infas, as documented in the methods report by Weitz and Jesske (2019). This data collection was motivated by the approach proposed by Schierholz et al. (2018), who introduced interactive occupation coding during interviews to mitigate errors arising from ambiguous respondent descriptions, missing information, and heterogeneity in coding practices. The specific coding instrument evaluated in the study was subsequently developed into a standardized software toolbox presented by Simson et al. (2023). The tool uses a machine learning algorithm to suggest a short list of occupational job descriptions from which the respondent selects the most appropriate match; this selection is then internally mapped to standardized occupational codes.

For each item, occupational information was coded by two professional human raters and, additionally, by the automated tool. This dual coding process was applied to both KldB and ISCO classification systems. For the tool output, both a preliminary code and a variant incorporating the requirement level (*Anforderungsniveau*) were available and respected in the further analyses. Tool flags regarding the requirement level are handled differently depending on the classification system: for KldB, both “requirement level not necessary” and “no code available for the chosen requirement level” are replaced by the preliminary tool code. For ISCO, however, only the former is replaced, while the latter is treated as missing. This distinction is made because ISCO-08 is more directly linked to qualification and skill levels through its major group structure, and a fallback would risk introducing systematic misclassification. This multiplicity of potentially conflicting labels motivates modeling the observed codes as noisy measurements of a latent true occupation.

Not all sources are observed for every item. Hence, missingness is an inherent part of the data setting and handled explicitly during data preparation.

The preparation pipeline pursues two objectives: first, to harmonize the code fields such that each source is represented in a comparable format, and second, to construct a controlled reduction to a common analysis level that is useful. The relevant variables are first selected from the full dataset and stored in a compact working table. All code fields are treated as text since some entries may contain status messages or multiple candidate codes rather than a single numeric code. A key cleaning step concerns tool-based fields that sometimes contain a flag instead of a code, for example, indicating that the requirement level is not necessary. In these cases, the corresponding preliminary tool code is used as a fallback so that the cleaned tool field contains a valid code whenever possible.

In a subset of cases, a single field contains multiple candidate codes, typically separated by commas or semicolons. Because these alternatives are not provided as an ordered list but rather as several plausible candidates, exactly one code per field is selected for the analysis. This selection is performed at random among valid candidate codes, and fixed random seeds are used to ensure reproducibility across runs. Only codes with the expected length are considered valid (five digits for KldB and four digits for ISCO); negative or non numeric entries are discarded.

For descriptive analyses, a coarse aggregation level is constructed by extracting the first

digit of the selected code. This first digit serves as an indicator of the major group and is stored separately for each source, that is, for the tool and the two human raters, as well as separately for KldB and ISCO. Missing values are standardized by converting empty strings and textual NA entries to proper missing values. Rows without any information in the considered fields are removed. Finally, each item is assigned a unique `item_id` to provide a stable reference for linking descriptive summaries and subsequent model based analyses.

3.3 Descriptive overview and inter rater agreement

Each item can be coded by an automated tool and by two professional raters. Because not all sources are observed for every item, the effective sample size differs across comparisons. In particular, the manual rater comparison is based on more items than comparisons involving the tool. For KldB, manual–manual agreement is computed on $n = 1322$ complete cases, whereas tool–manual and three-way agreement rely on $n = 932$ items. For ISCO, the manual–manual comparison uses $n = 1326$ items (almost the same size as for KldB), while tool–manual comparisons and the three-way comparison rely on only about $n \approx 840$ items.

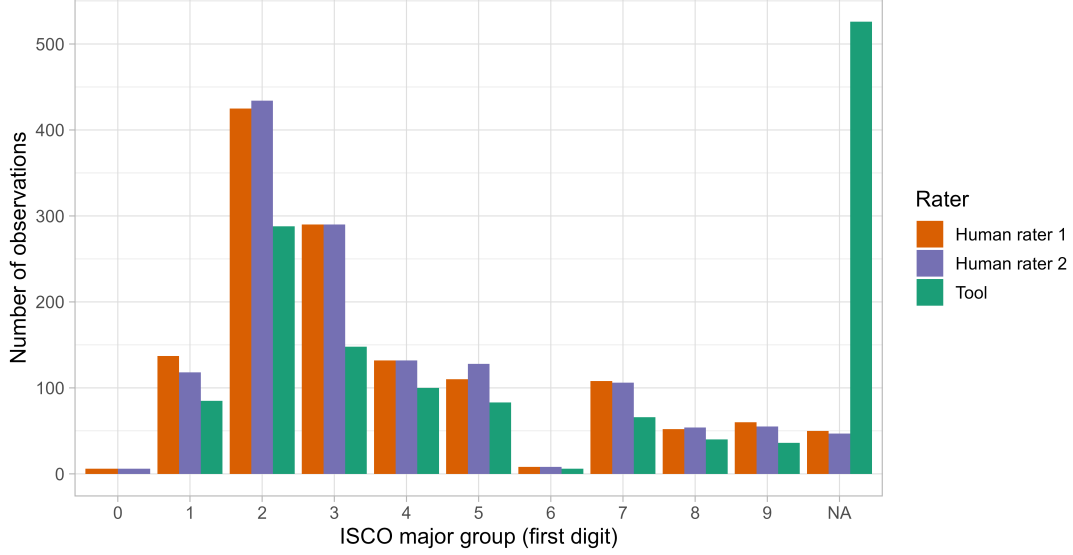
For the descriptive plots, the original KldB 2010 and ISCO-08 codes are reduced to their major groups, defined as the first digit of the respective code. This yields a coarse but comparable view of how items are distributed across occupational groups, and it allows agreement to be summarized in a transparent way.

Figure 2 shows the resulting major group distributions by source. In both classification schemes, the distributions are clearly imbalanced: a small number of major groups accounts for a large share of the observed codes, while several groups occur only rarely. For ISCO, the distribution is dominated by major groups “Professionals” (digit 2) and “Technicians and Associate Professionals” (digit 3), while groups such as 0 and 6 occur only sparsely. For KldB, the major groups “Business organization, accounting, law and administration” (7) and “Health, social services, teaching and education” (8) dominate across sources, followed by group 2, whereas groups such as 0 and 1 are comparatively rare. To ensure easier readability, in the following sections only the numbered codes are referred to. It is important to recall that the interpretation of the numbers is not consistent throughout the schemes. Across both schemes, the tool output also exhibits a visibly larger fraction of missing values than the manual ratings, consistent with the presence of non-code status messages in tool fields that are treated as missing during preprocessing.

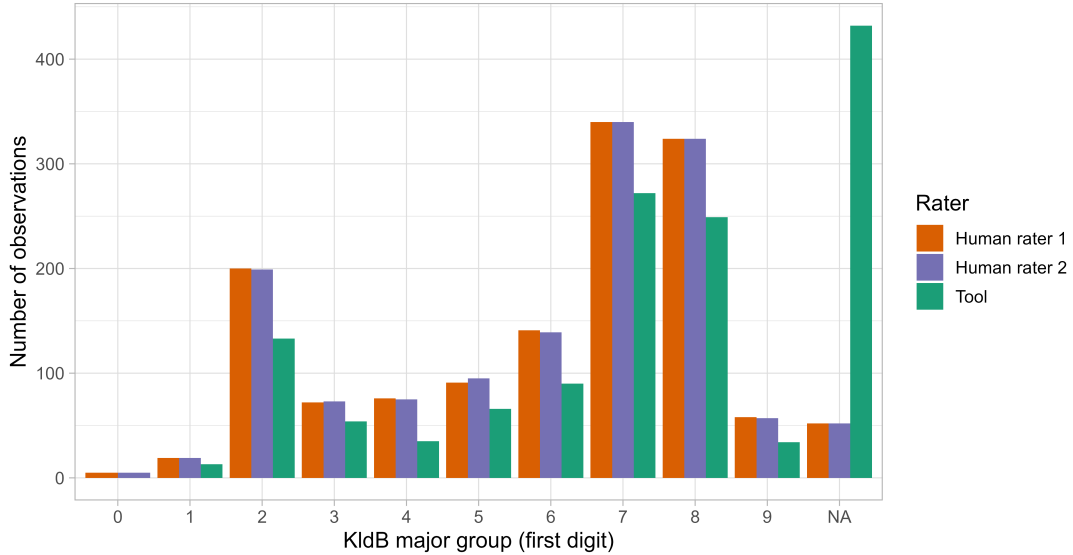
Agreement patterns differ strongly between KldB and ISCO, see Figure 3. For ISCO, mismatch rates are consistently higher across all comparisons. Even between the two professional raters, the mismatch rate is 14.5%, while tool–manual mismatch is between 23% and 29%. When requiring agreement across all three sources, the mismatch rate increases to 31.1%. The fact that the three-way mismatch rate exceeds both tool–manual mismatch rates indicates that disagreement is not confined to a single source but is more diffuse across all three coding outputs.

For KldB, the two professional raters disagree only rarely at the major group level (mis-

match 2.1%), whereas disagreement between the tool and the human raters is substantially higher (around 10%). Requiring all three sources to agree yields a three-way mismatch rate of 10.6%, which is close to the tool–manual mismatch rates. Taken together, this pattern suggests that most residual disagreement at this aggregation level is driven by deviations of the tool rather than by inconsistencies between the two human coders.



(a) ISCO: Distribution of major groups (first digit)



(b) KldB: Distribution of major groups (first digit)

Figure 2: Major group distributions by source after preprocessing and reduction to the first digit. Note that the digits refer to different occupational group definitions in KldB and ISCO and are therefore not comparable across classification schemes.

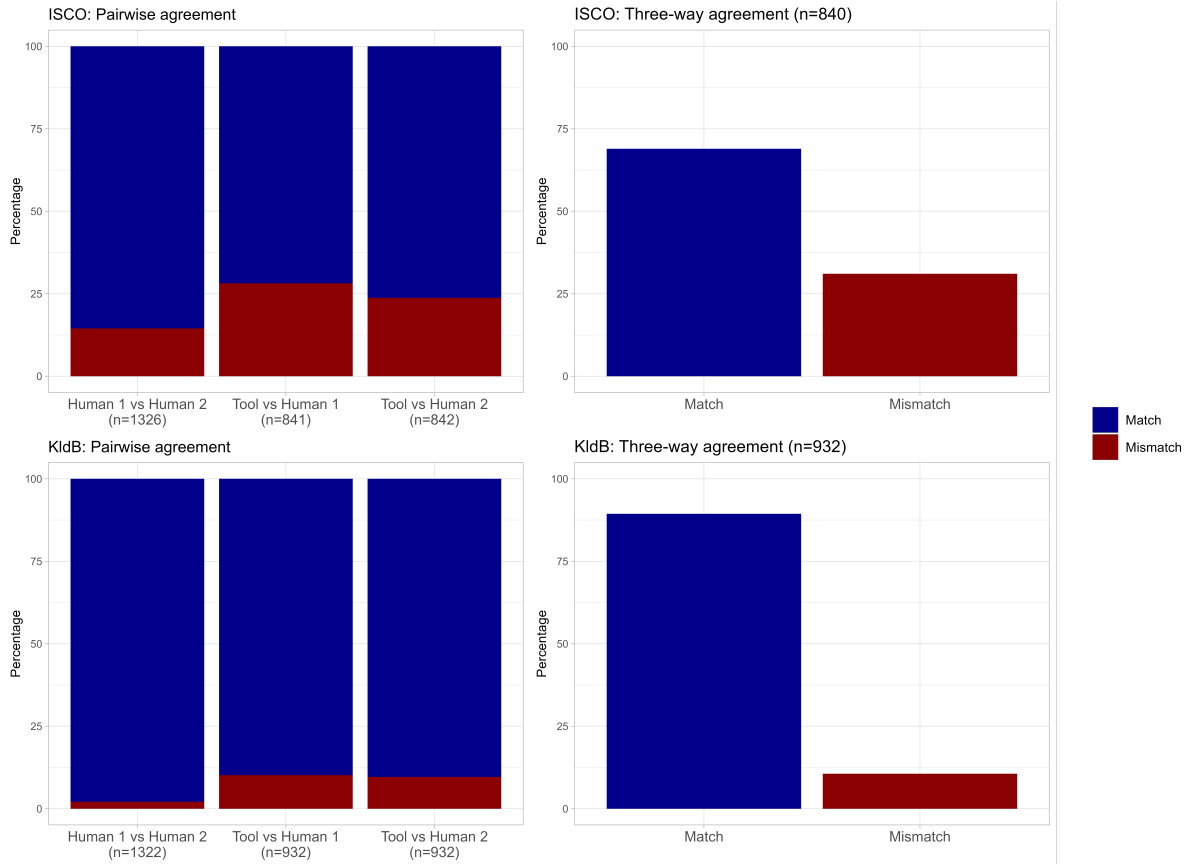


Figure 3: Inter-source agreement on the first digit for ISCO (top row) and KldB (bottom row), showing pairwise comparisons (left column) and three-way agreement (right column). Values are reported as percentages with sample sizes (n) indicated. Blue bars represent matches, red bars represent mismatches.

3.3.1 Data preparation for modeling

For the Bayesian noisy measurement model, the analysis starts with a binary setting as a pragmatic step to obtain a stable and estimable model fit in this dataset. Given the available sample size, this reduction is necessary to limit sparsity and obtain stable rater quality estimates. The binary specification serves as a baseline, from which extensions to more fine-grained category sets can be explored. The preparation proceeds in two steps and is carried out separately for KldB and ISCO:

1. *Restriction to the two most frequent categories.* For each classification system, only the two most common categories at the chosen aggregation level are retained. All subsequent modeling is performed on this two-class subset.
2. *Complete-case filtering within the two-category subset.* Only cases are kept for which all available ratings are complete and take values exclusively in the selected two categories. Any case in which at least one source assigns a category outside the two-class set is removed, so the modeled dataset contains no additional categories beyond the binary reduction.

This yields a clean two-class dataset in which each retained case can be consistently interpreted as belonging to one of the two focal categories, with missingness and out-of-scope labels removed by construction.

Observed agreement in the two-class subsets. To characterize the empirical disagreement induced by the binary reductions, agreement is summarized for the retained two-category subsets, and the substantive labels of the categories are stated explicitly.

For ISCO (2 versus 3), the two retained major groups are *Professionals* (2) and *Technicians and Associate Professionals* (3). Figure 4 summarizes observed agreement on this subset. The pairwise comparison (left panel) shows that the two human raters agree more often with each other than either agrees with the tool, and the tool is therefore a frequent driver of disagreement. The three-way summary (right panel) indicates that a non-negligible share of items (11.8%) exhibits at least one disagreement across the three sources. This higher level of disagreement makes the ISCO subset particularly informative for assessing rater-specific error patterns in the subsequent model fit.

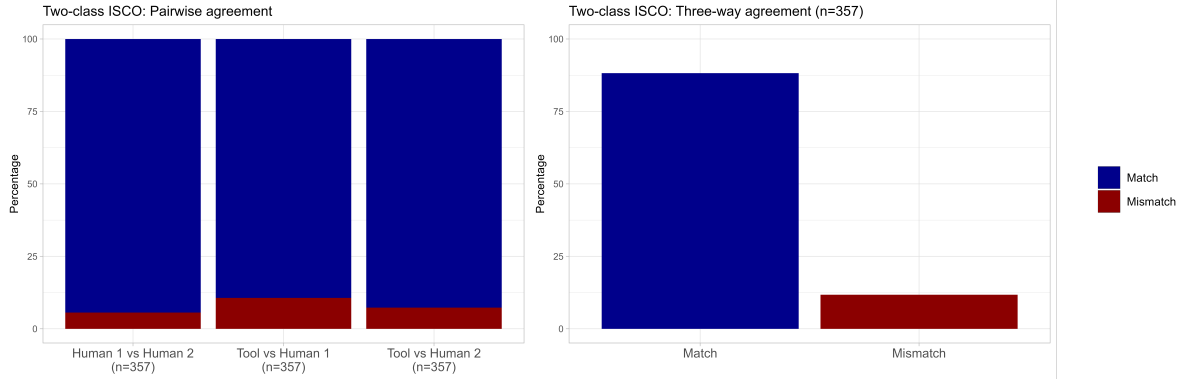


Figure 4: ISCO (Professionals vs Technicians and Associate Professionals): observed pairwise agreement (left) and three-way agreement (right) on the two-class subset.

For KldB (7 versus 8), the two retained major groups are *Business organisation, accounting, law and administration* (7) and *Health, social services, teaching and education* (8). The retained subset is nearly unanimous across all three sources. Out of 471 cases, 225 are labeled as 7 by all three mechanisms and 242 as 8 by all three mechanisms. Only 4 cases show disagreement, and these disagreements are all of the same type: both human raters assign class 8, while the tool assigns class 7. No cases exhibit disagreement between the two human raters, and no cases show the opposite direction (human 7 with tool 8). While this implies that the ISCO subset contains substantially more disagreement and is therefore more informative for global rater-quality estimation, the KldB subset remains substantively interesting because the four discordant cases represent a small, clearly characterized set of tool-only deviations that can be inspected qualitatively as potential systematic boundary errors.

Table 1: KldB (7 vs 8): observed three-way rating patterns after filtering.

Observed pattern (Human 1, Human 2, Tool)	Count	Share (%)
(7, 7, 7)	225	47.8
(8, 8, 8)	242	51.4
(8, 8, 7)	4	0.8
Total	471	100.0

4 Model application and Results

4.1 Prior specification and hyperparameter choices

Recall: The Bayesian noisy measurement model with two latent categories is used (major group 2 vs 3 for ISCO and major group 7 vs 8 for KldB). Let C_i denote the latent true class of item i , and let $p = (p_1, p_2)$ be the latent class proportions. For each rater $k \in \{1, \dots, K\}$, a class-conditional confusion matrix $\Pi^{(k)}$ is modeled with rows indexed by the true class and columns by the observed class. The entry $\pi_{j\ell}^{(k)}$ denotes the probability that rater k reports class ℓ when the true class is j .

$K = 3$ denotes the number of raters (two human coders and one tool in this application). J denotes the number of latent categories. In the main results reported below, $J = 2$ (the two-category variant). The three-category extension corresponds to $J = 3$ and is described in Section 4.5.

Prior on class proportions. The prevalence vector p receives a Dirichlet prior

$$p \sim \text{Dirichlet}(\alpha).$$

As a baseline, a weakly informative $\alpha = (1, 1)$ prior is used, which is uniform over the simplex and therefore does not privilege either class a priori. In the prior sensitivity analysis (ISCO only), a more informative alternative $\alpha = (300, 200)$ is additionally considered to encode a prior belief that major group 2 is more common than major group 3. Under this specification, the prior mean for Dirichlet (Bishop, 2006) is

$$\mathbb{E}[p_{\text{ISCO}=2}] = \frac{300}{300 + 200} = 0.6, \quad \mathbb{E}[p_{\text{ISCO}=3}] = \frac{200}{300 + 200} = 0.4.$$

This specification was chosen to mimic the order of magnitude and the direction of the imbalance observed in the descriptive analysis (see Figure 2). By setting the concentration parameter sum to 500, the prior information is given a weight roughly comparable to the observed sample size. This creates a strong informative prior that anchors the class ratios around the empirically observed values, serving as a robust sensitivity check against the weakly informative baseline.

Prior on confusion matrices. Each row of each confusion matrix is assigned a Dirichlet prior. For the two-class case, each row is a length-2 simplex. For each rater k and true class j , it is set that

$$\pi_{j\cdot}^{(k)} \sim \text{Dirichlet}(\beta_{j\cdot}),$$

and two specifications for the Dirichlet parameter vector $\beta_{j\cdot}$ are compared:

- **Uniform:** $\beta_{j\cdot} = (1, 1)$, implying a prior mean 0.5 on the diagonal and very weak concentration.
- **Diagonal-favoring:** $\beta_{j\cdot} = (5, 1)$ for $j = 1$ and $\beta_{j\cdot} = (1, 5)$ for $j = 2$, which implies

$$\mathbb{E}\left[\pi_{jj}^{(k)}\right] = \frac{5}{5 + 1} = 83.3\% \quad \text{and} \quad \mathbb{E}\left[\pi_{j\ell}^{(k)}\right] = \frac{1}{6} = 16.7\% \quad (\ell \neq j).$$

The latter choice encodes the mild, practically motivated belief that a rater is more likely correct than incorrect: under $\beta_j = (5, 1)$ the prior mean of the diagonal entry (i.e., the class-conditional probability of reporting the correct label given true class j) is $\mathbb{E}[\pi_{jj}^{(k)}]$, so the prior centers each confusion-matrix row at roughly 83% correct classification while still allowing substantial deviation when the data suggest otherwise.

4.2 Results: 2-category variant

ISCO is presented first as the primary reference case (larger discrepancies and hence more informative patterns), followed by a shorter KldB summary. Unless stated otherwise, ISCO results in this section refer to the reference specification $\alpha = (1, 1)$, uniform β , meaning the baseline model uses uniform priors.

For KldB, results refer to the same baseline specification with $\alpha = (1, 1)$ and a uniform confusion-matrix prior $\beta = (1, 1)$.

4.2.1 ISCO: posterior summaries of latent classes and class proportions

Table 2 reports posterior summaries for the latent class proportions under the reference specification $\alpha = (1, 1)$ and uniform β_j . The posterior mean indicates that major group 2 is more prevalent than major group 3, which aligns with the observed, reported data.

Table 2: ISCO: posterior summaries of class proportions p for the two-category variant ($\alpha = (1, 1)$, uniform β).

Parameter	Mean	SD	5%	95%
p_1 (class 2)	0.726	0.0240	0.685	0.765
p_2 (class 3)	0.274	0.0240	0.235	0.315

The columns 5% and 95% report the 0.05 and 0.95 posterior quantiles, yielding a 90% credible interval for each component of p .

For a compact visual summary, Figure 5 shows the posterior mean of p . This posterior distribution is broadly consistent with the marginal distribution of raw annotations shown in Section 3.3.

4.2.2 ISCO: rater confusion patterns

The posterior mean confusion matrices under the baseline model ($\alpha = (1, 1)$, uniform β) indicate practically important differences between the two human raters and the tool. Table 3 reports the posterior means for each rater, with rows corresponding to the latent true class and columns to the observed class.

A first conclusion is that both human raters place most of the probability mass on the diagonal, indicating that they consistently reproduce the latent class. Human rater 2 approaches a near-deterministic mapping: conditional on true class 2, the probability of reporting 2 is about 0.987, and conditional on true class 3, it is about 0.979. This indicates

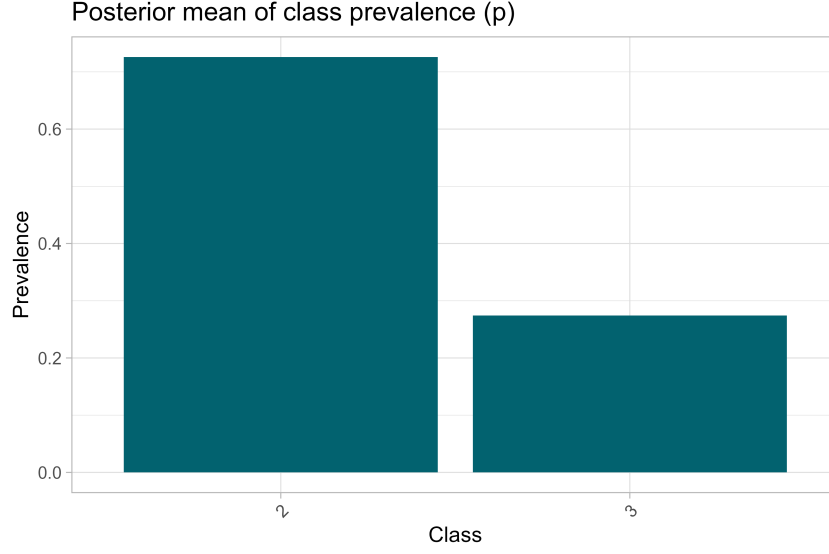


Figure 5: ISCO: posterior mean of class proportions for the two-category variant ($\alpha = (1, 1)$, uniform β).

a high degree of precision, characterized by negligible and nearly symmetric residual error probabilities.

Human rater 1 performs clearly above chance but is less sharp than rater 2. The off-diagonal probabilities are 0.047 for true 2 being recorded as 3, and 0.068 for true 3 being recorded as 2. This reveals a directional asymmetry: the probability of misclassifying true class 3 as class 2 exceeds that of the reverse error, suggesting that identifying class 3 is comparatively more challenging for this rater.

The Machine Learning-powered tool shows the same direction of asymmetry but is substantially amplified. For true class 2, the tool looks very similar to human rater 1, with an off-diagonal mass of about 0.047 (predicting 3 when the true class is 2). However, for true class 3, the tool appears to perform markedly weaker: the posterior mean probability of predicting 2 given true class 3 is about 0.130. This is roughly twice the error rate of human rater 1 in the same row (0.068) and more than six times the error rate of human rater 2 (0.021). Hence, the primary weakness of the tool is not a general lack of accuracy; rather, it is a class-specific tendency to collapse ambiguous cases into class 2 when the latent class is 3.

Table 3: ISCO: posterior mean confusion matrices (rows: true class, columns: observed class) under the baseline model ($\alpha = (1, 1)$, uniform β).

Human rater 1			Human rater 2			Tool		
	obs 2	obs 3		obs 2	obs 3		obs 2	obs 3
true 2	0.953	0.047	true 2	0.987	0.013	true 2	0.953	0.047
true 3	0.068	0.932	true 3	0.021	0.979	true 3	0.130	0.870

Credible intervals for diagonal entries and rater sharpness. To quantify how precisely these differences are estimated, Table 4 reports 90% credible intervals (5% to 95%) for the diagonal entries of each rater’s confusion matrix. These intervals confirm that all three sources are well above chance on both classes, but they also show that the tool is substantially less sharp on true class 3. While the upper bound of the tool’s interval (0.924) overlaps with the lower range of human rater 1, indicating that comparable performance is theoretically possible within the posterior uncertainty, the contrast to human rater 2 is strict. Crucially, the lower bound (5% quantile) for the tool drops to roughly 0.809, whereas both human raters remain reliably above 0.88 even in the conservative tail of the posterior.

Table 4: ISCO: posterior summaries (Mean, SD, 5%, 95%) for diagonal entries $\pi_{jj}^{(k)}$ under the baseline model ($\alpha = (1, 1)$, uniform β).

Rater	Diagonal entry	Mean	SD	5%	95%
Human rater 1	$\pi_{2,2}^{(1)}$	0.953	0.0135	0.929	0.973
Human rater 1	$\pi_{3,3}^{(1)}$	0.932	0.0274	0.882	0.971
Human rater 2	$\pi_{2,2}^{(2)}$	0.987	0.0084	0.971	0.998
Human rater 2	$\pi_{3,3}^{(2)}$	0.979	0.0166	0.947	0.998
Tool	$\pi_{2,2}^{(3)}$	0.953	0.0131	0.930	0.972
Tool	$\pi_{3,3}^{(3)}$	0.870	0.0350	0.809	0.924

Weighted diagonal mass. This pattern has a direct applied interpretation. Major group 2 is also the more prevalent class in ISCO under the model (Table 2), so a systematic bias toward predicting 2 is particularly consequential. It can appear harmless in overall agreement rates, but it specifically undermines the detection of class 3. In operational terms, if class 3 is the category where correct identification matters, the tool seems to be the least reliable source precisely in the row that matters most.

To quantify rater performance in a single number that accounts for class imbalance, the prevalence-weighted diagonal mass is used

$$D_k = \sum_{j \in \{2,3\}} p_j \pi_{jj}^{(k)}.$$

Under the baseline model ($\alpha = (1, 1)$, uniform β), $D_k \approx 0.947$ is obtained for human rater 1, 0.985 for human rater 2, and 0.930 for the tool. Because the tool closely matches human rater 1 for true class 2 but is much worse for true class 3, its overall D_k ends up below both humans, even though one row is comparable. This is the core substantive result for ISCO: the tool is not uniformly bad; it fails in a specific direction, namely confusing true 3 as 2, and this direction is large enough to create a stable performance gap relative to human rater 2, and also a lower overall D_k than human rater 1.

Table 5: ISCO: rater-level weighted diagonal mass D_k for the baseline model ($\alpha = (1, 1)$, uniform β).

Rater	D_k	$1 - D_k$
Human rater 1	0.947	0.053
Human rater 2	0.985	0.015
Tool	0.930	0.070

4.2.3 KldB: posterior summaries of latent classes and class proportions

For KldB, the same two-category Dawid–Skene model is fitted, restricting the first digit to major groups 7 and 8. All KldB results reported in this section use the baseline specification with a uniform prior on class proportions and uniform row priors on the confusion matrices, referred to as baseline model. Concretely, the prevalence vector p is assigned $\alpha = (1, 1)$ and each confusion-matrix row receives $\beta_j = (1, 1)$. This baseline choice avoids injecting prior information into a setting where the observed three-way ratings are already almost perfectly consistent (only 4 disagreements in the retained subset).

 Table 6: KldB: posterior summaries of class proportions p for the two-category variant ($\alpha = (1, 1)$, uniform β).

Parameter	Mean	SD	5%	95%
p_7 (class 7)	0.478	0.0229	0.441	0.516
p_8 (class 8)	0.522	0.0229	0.484	0.559

The columns 5% and 95% report the 0.05 and 0.95 posterior quantiles, yielding a 90% credible interval for each component of p . Under the baseline specification, the posterior for the latent class proportions is close to balanced, with a slight tilt toward class 8. An overlap of the credible intervals indicates that values with $p_7 > p_8$ remain plausible.

For a compact visual summary, Figure 6 again shows the posterior mean of p .

4.2.4 KldB: rater confusion patterns

Table 7 reports the posterior mean confusion matrices for the baseline model ($\alpha = (1, 1)$, uniform β). Rows correspond to the latent true class (7 or 8) and columns correspond to the observed class recorded by the respective source.

 Table 7: KldB: posterior mean confusion matrices (rows: true class, columns: observed class) for the baseline model ($\alpha = (1, 1)$, uniform β).

Human rater 1			Human rater 2			Tool		
	obs 7	obs 8		obs 7	obs 8		obs 7	obs 8
true 7	0.996	0.004	true 7	0.996	0.004	true 7	0.996	0.004
true 8	0.004	0.996	true 8	0.004	0.996	true 8	0.020	0.980

According to the model, both human raters are almost perfectly consistent with the latent classification in both rows, with off-diagonal probabilities around 0.004 in each direction.

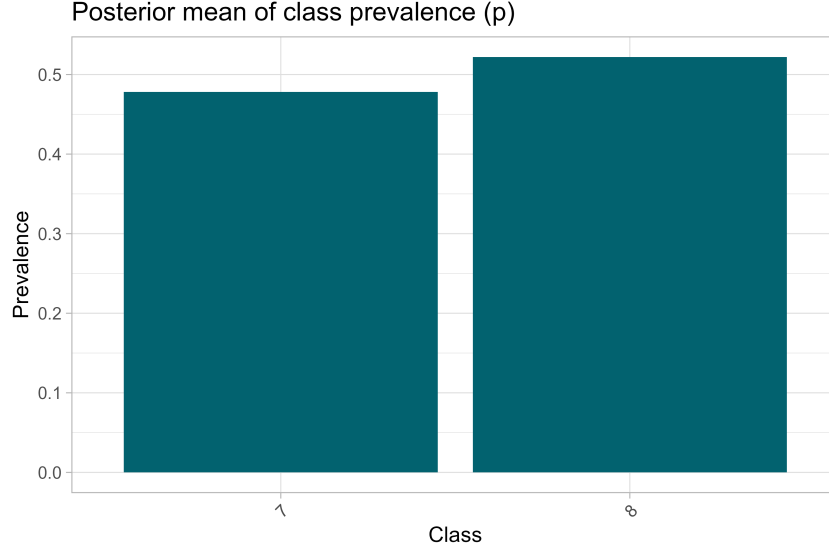


Figure 6: KldB: posterior mean of class proportions for the two-category variant ($\alpha = (1, 1)$, uniform β).

The tool is similarly strong for true class 7 but shows a class-specific weakness for true class 8. Conditional on latent class 8, the posterior mean probability that the tool reports 7 is about 0.020, which is substantially larger than the corresponding human off-diagonal probabilities, even though it remains small in absolute terms.

To make the uncertainty behind this conclusion explicit, Table 8 reports 90% credible intervals for the most relevant off-diagonal cells. All misclassification probabilities concentrate close to zero, with the tool’s probability of mapping true 8 into 7 remaining clearly separated from the human range.

Table 8: KldB: posterior means and 90% credible intervals for selected off-diagonal confusion probabilities under the baseline model ($\alpha = (1, 1)$, uniform β).

Cell (true $\rightarrow$ observed)	Mean	SD	5%	95%
Human 1: 8 $\rightarrow$ 7	0.0041	0.0040	0.0002	0.0121
Human 2: 8 $\rightarrow$ 7	0.0040	0.0041	0.0002	0.0121
Tool: 8 $\rightarrow$ 7	0.0201	0.0089	0.0080	0.0363
Human 1: 7 $\rightarrow$ 8	0.0044	0.0045	0.0002	0.0134
Human 2: 7 $\rightarrow$ 8	0.0044	0.0044	0.0002	0.0132
Tool: 7 $\rightarrow$ 8	0.0044	0.0043	0.0002	0.0131

To summarize rater performance in a single number that accounts for prevalence, the weighted diagonal mass is again reported,

$$D_k = \sum_{j \in \{7,8\}} p_j \pi_{jj}^{(k)}.$$

Under $\alpha = (1, 1)$ and uniform β , $D_k \approx 0.996$ for both human raters and 0.987 for the tool,

reflecting that the only practically relevant gap is concentrated in the true-8 row rather than being uniform across classes.

Table 9: KldB: rater-level weighted diagonal and off-diagonal mass for the baseline model ($\alpha = (1, 1)$, uniform β).

Rater	D_k	$1 - D_k$
Human rater 1	0.9958	0.0042
Human rater 2	0.9958	0.0042
Tool	0.9874	0.0126

Overall, the KldB two-category reduction (7 vs 8) is so clean that posterior uncertainty at the item level is essentially absent: the 95% credible set for C_i is a singleton for every item in the retained subset. The model-based conclusions are therefore dominated by the near-deterministic agreement structure, but the remaining signal is still interpretable and systematic: the tool exhibits a small, directional tendency to map true 8 into 7.

4.3 Convergence and diagnostics

4.3.1 HMC/NUTS in brief and Diagnostics used

Posterior inference in Stan is obtained via Hamiltonian Monte Carlo (HMC) using the No-U-Turn Sampler (NUTS) (Stan Development Team, 2024). Intuitively, the algorithm proposes new parameter values by using gradient information to move efficiently through regions of high posterior density, rather than taking small random steps. NUTS automates the choice of step length, allowing the chains to explore the posterior reliably with little manual tuning.

The sampling procedure is configured with a small set of practical parameters:

- **chains**: number of independent Markov chains. Multiple chains allow convergence checks via between-chain vs. within-chain comparisons.
- **parallel_chains**: number of chains run in parallel, affecting runtime but not the target distribution.
- **iter_warmup**: warmup (also referred to as “Burn-in”) iterations used for adaptation (step size and mass matrix). Warmup draws are discarded from posterior summaries.
- **iter_sampling**: sampling iterations retained for posterior inference.
- **adapt_delta**: target acceptance probability. Higher values typically reduce divergences at the cost of more conservative (smaller) step sizes and increased runtime.
- **max_treedepth** (if set): maximum depth of the NUTS tree, limiting trajectory length and preventing excessive computation per iteration.

4.3.2 Empirical convergence assessment

Sampling quality is assessed using standard diagnostics as provided by Stan and visualized with the interactive R package `shinystan` (Gabry and Veen, 2025). The key acceptance criteria are:

- **Split- $\hat{R}$:** values close to 1 indicate agreement across chains. Concretely, split- $\hat{R}$ compares between-chain variation to within-chain variation. If chains explore the same posterior region, both variations match and split- $\hat{R} \approx 1$. Values above 1.01 are inspected, and values above 1.1 indicate clear non-convergence (typically meaning at least one chain has not mixed into the same region as the others) (Stan Development Team, 2024). It is therefore considered the most important convergence assessment measure.
- **Effective sample size (ESS):** Stan reports two variants, `ess_bulk` for the central posterior mass and `ess_tail` for tail behavior (relevant for quantiles and interval bounds). In `shinystan`, the plot n_{eff}/N displays the effective sample size relative to the total number of post-warmup draws N , so values closer to 1 indicate weak autocorrelation. As a pragmatic warning rule, the default `shinystan` threshold is followed and parameters with $n_{\text{eff}}/N < 0.10$ are flagged. If less than 10% of draws are effectively independent, posterior means and, in particular, tail summaries become noticeably less stable unless N is very large.
- **Divergences and treedepth:** divergences indicate that HMC could not follow the posterior geometry reliably (potentially biasing exploration). Frequent maximum-treedepth hits suggest that the chosen trajectory length was often insufficient for efficient exploration per iteration.
- **Autocorrelation:** Autocorrelation plots for key quantities are inspected. Rapid decay of autocorrelation with increasing lag indicates efficient exploration and supports the interpretation that the effective sample size is sufficiently large. Persistently high autocorrelation suggests slow mixing and unstable summaries.

4.3.3 Applied robustness checks

Beyond the reference fit, a robustness check was performed in which individual sampling settings were varied one at a time around the reference configuration. This check was carried out exemplarily for ISCO (two-category model with priors $\alpha = (1, 1)$, uniform β), because ISCO exhibits a more informative disagreement structure and is therefore a more demanding case for the sampler.

Exemplar diagnostic snapshots. Appendix A.3 shows three representative diagnostic views for the reference ISCO fit. First, the sampler summary indicates stable HMC behavior: the divergent transition rate is zero, the average tree depth remains low, and the acceptance statistic is high, which together suggest that NUTS explores the posterior without repeatedly running into problematic geometry. Second, the autocorrelation plot for the log-posterior decays quickly with lag in all chains, which indicates good mixing and supports the effective sample size reported by Stan. Third, the split- $\hat{R}$ distribution is

tightly concentrated near 1, and **shinystan** reports no parameters exceeding the chosen warning threshold, which supports the use of posterior summaries from this fit.

What was varied Table 10 summarizes the one-at-a-time robustness checks around the reference configuration. Across these checks, convergence diagnostics were stable for essentially all single-setting changes. The only configuration that produced clear non-convergence signals was the increase in the number of chains, where $\text{split-}\hat{R} > 1.1$ appeared for multiple components of p and $\Pi^{(k)}$. This behavior is consistent with weak identification: with more chains, it becomes more likely that at least one chain explores a different posterior region, inflating between-chain variation.

Table 10: One-at-a-time robustness checks around the reference sampler configuration (ISCO, two-category). Only increasing the number of chains produced $\text{split-}\hat{R} > 1.1$ in this check.

Setting	Lower	Reference	Higher	Converged?
<code>chains</code>	2	4	6	No (higher)
<code>parallel_chains</code>	2	4	4	Yes
<code>iter_warmup</code>	500	1000	2000	Yes
<code>iter_sampling</code>	2000	4000	8000	Yes
<code>refresh</code>	100	200	400	Yes
<code>adapt_delta</code>	0.90	0.99	0.999	Yes

For every model reported in this thesis, convergence was explicitly checked using $\text{split-}\hat{R}$ and effective sample sizes, complemented by diagnostic plots where needed. Only fits that pass these checks are used for interpretation. While many further combinations of settings that could lead to divergence could be explored, an exhaustive grid search over sampler configurations is out of scope here.

The sensitivity analysis reveals that among converged fits, posterior estimates for class proportions and rater-specific summaries exhibit only marginal variation. Consequently, the substantive findings are robust to variations in the sampler’s hyperparameter settings.

4.4 Prior sensitivity: influence of alpha and beta

4.4.1 Sensitivity design

To determine to what extent the choice of priors α and β influences the model results, the prior on class proportions (α) and the prior on confusion matrices ($\beta_{j\cdot}$) are varied in a small grid. This prior sensitivity analysis was carried out for ISCO only. Concretely, four models are compared for ISCO:

$$\alpha \in \{(1, 1), (300, 200)\}, \quad \beta_{j\cdot} \in \{\text{uniform, diagonal-favoring (5)}\}.$$

For each ISCO model, (i) posterior mean class proportions, (ii) weighted diagonal mass per rater, and (iii) item-level uncertainty metrics are summarized.

4.4.2 Impact on latent class prevalence

For ISCO (restricted to categories 2 and 3), varying the Dirichlet hyperparameter α has a clear effect on the posterior mean prevalence vector p . Under the weakly informative baseline $\alpha = (1, 1)$, the posterior mean prevalence of class 2 is about 0.726 (with class 3 at about 0.274), and this is essentially unchanged when switching β . In contrast, under the strongly informative alternative $\alpha = (300, 200)$, the posterior mean prevalence shifts markedly toward the prior mean $300/(300 + 200) = 0.6$, yielding a posterior mean of about 0.652 for class 2 and 0.348 for class 3. Hence, the change in α moves the posterior mean for p_2 by roughly 7.4 percentage points, which is no longer negligible.

It may seem counter-intuitive that anchoring the prior expectation at 0.6 for class 2 leads to a posterior estimate ($p_2 \approx 0.65$) that is lower than the uninformative baseline ($p_2 \approx 0.73$). This occurs because the observed data exhibits an even stronger imbalance than the prior. Crucially, the high magnitude of the hyperparameters ($\alpha = (300, 200)$) assigns this prior a weight equivalent to 500 pseudo-observations, which exceeds the actual sample size of the subset ($n = 357$). Consequently, the informative specification acts as a strong conservative force ('shrinkage'), pulling the posterior estimate away from the extreme data likelihood and towards the more moderate 60:40 prior expectation.

Figure 7 visualizes this result. The two $\alpha = (1, 1)$ bars are almost identical, whereas both $\alpha = (300, 200)$ bars are clearly separated from them, indicating substantial shrinkage of the inferred class imbalance toward the informative prior.

4.4.3 Impact on confusion structure and uncertainty

The hyperparameter β regularizes the rows of the confusion matrices $\pi^{(k)}$ and therefore directly targets diagonal concentration, that is, the probability of assigning the correct class conditional on the latent true class. In the present application, β has only a minor effect on the fitted rater behavior, while α induces noticeable changes in prevalence-weighted summaries.

First, rater performance summarized by the weighted diagonal mass remains very stable with respect to β within each α setting. Under $\alpha = (1, 1)$, switching from uniform to

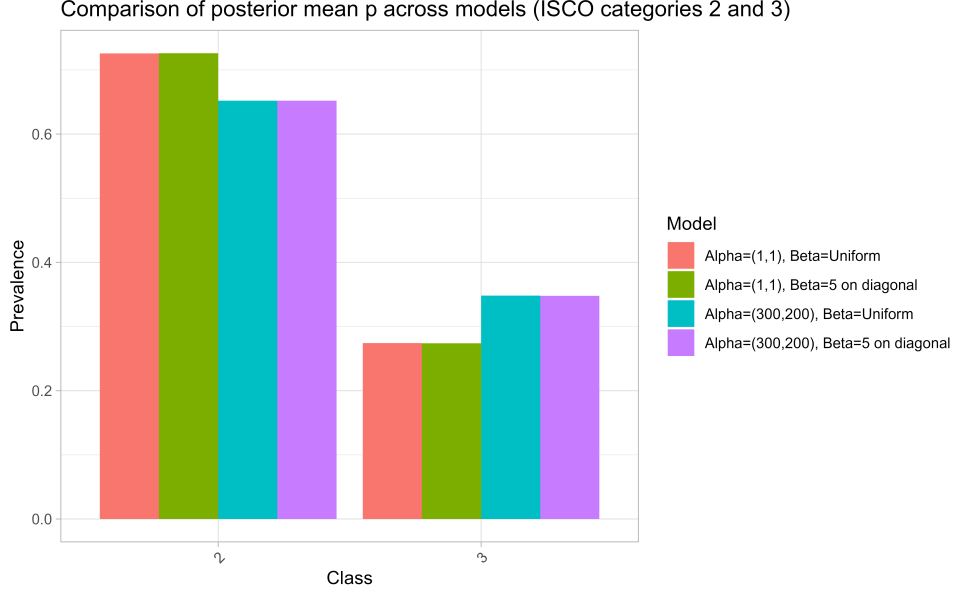


Figure 7: Posterior mean prevalence p for ISCO classes 2 and 3 across the four (α, β) configurations.

diagonal-favoring β changes the weighted diagonal mass by about +0.0013 for human rater 1 (from about 0.9474 to 0.9487), by about +0.0002 for human rater 2 (from about 0.9846 to 0.9848), and by about +0.0019 for the tool (from about 0.9303 to 0.9322). Under $\alpha = (300, 200)$, the corresponding β effect is again small (about +0.0015 for human rater 1, +0.0005 for human rater 2, and +0.0025 for the tool). This pattern is shown in Figure 8.

In contrast, changing α has a clearly visible impact on the tool’s weighted diagonal mass: it drops from about 0.930 under $\alpha = (1, 1)$ to about 0.923 under $\alpha = (300, 200)$ for uniform β , and from about 0.932 to about 0.925 under diagonal-favoring β . This is a consequence of the definition $D_k = \sum_j p_j \pi_{jj}^{(k)}$: the informative α increases the posterior weight on class 3, and the tool is the weakest source in the true-3 row, so the prevalence-weighted accuracy decreases even if the row-wise confusion probabilities remain similar.

Second, the most direct view of the β effect is obtained by comparing the diagonal elements $\pi_{jj}^{(k)}$ by true class and rater. Figure 9 shows a slight upward shift of the diagonal entries $\pi_{jj}^{(k)}$ under diagonal-favoring β , but the relative pattern across sources and classes remains unchanged, in particular the tool remains least accurate in the true-3 row.

Finally, item-level uncertainty remains very low and nearly unchanged across all four configurations, it increases slightly under the informative α . The mean posterior entropy rises from about 0.0168 to 0.0172 under $\alpha = (1, 1)$ to about 0.0187 to 0.0194 under $\alpha = (300, 200)$. Detailed uncertainty measures and their interpretation are discussed in Section 5.1.

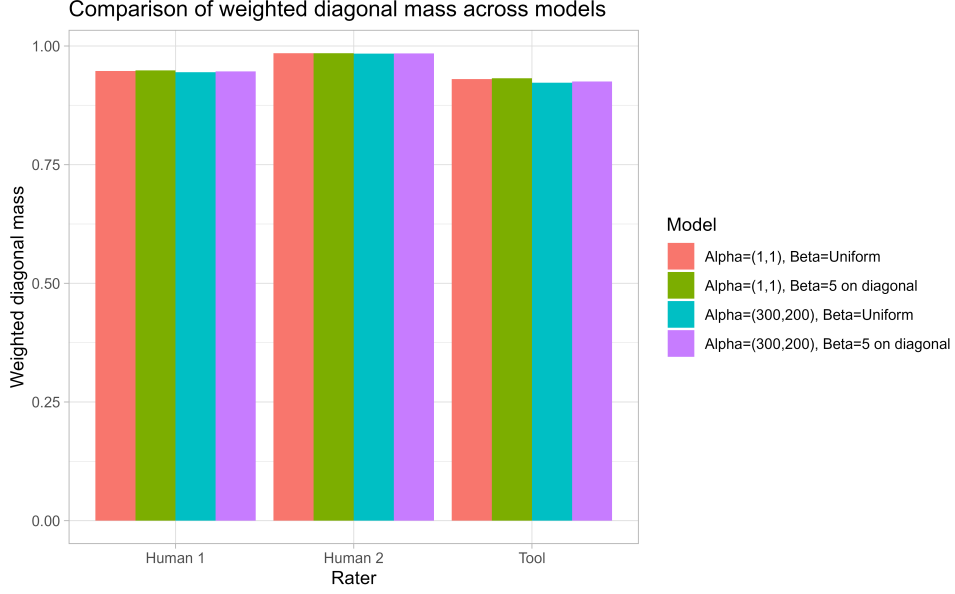


Figure 8: Weighted diagonal mass across the four (α, β) configurations.

4.5 Extension: More-category variant

The main results above focus on binary versions of the model because this regime is the most stable given the available information per category. This section extends the Bayesian Dawid and Skene model from $J = 2$ to $J = 3$ categories and illustrates why prior regularization becomes much more important once J increases. In particular, stable sampling without label switching required a sufficiently strong diagonal-favoring prior on the confusion matrices.

4.5.1 Model extension and parameter growth

Recall that J is the number of latent classes and K the number of raters. The model parameters are the class proportions $p \in \Delta^{J-1}$ and, for each rater k , a $J \times J$ confusion matrix $\Pi^{(k)}$, with the constraints that p lies on the simplex and each row of $\Pi^{(k)}$ sums to one. Counting free parameters under the simplex constraints yields

$$\#\text{free}(p) = J - 1,$$

since the normalization constraint $\sum_{j=1}^J p_j = 1$ reduces the degrees of freedom by one.

For each rater k , $\Pi^{(k)}$ is a $J \times J$ row-stochastic matrix. Each row has $J - 1$ free entries, hence

$$\#\text{free}(\Pi^{(k)}) = J(J - 1).$$

With K raters, the total number of free parameters is therefore

$$\#\text{free total} = \#\text{free}(p) + \sum_{k=1}^K \#\text{free}(\Pi^{(k)}) = (J - 1) + KJ(J - 1) = (J - 1)(1 + KJ).$$

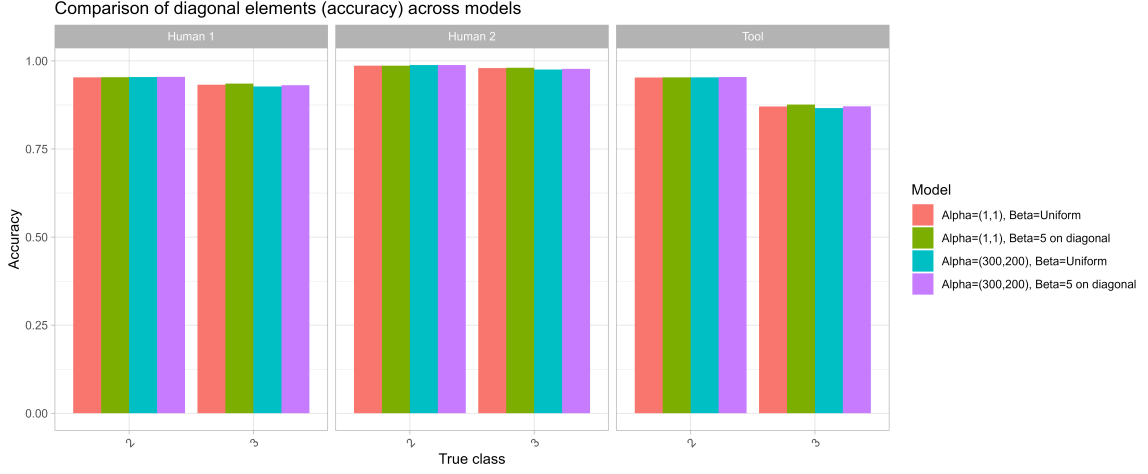


Figure 9: Diagonal elements $\pi_{jj}^{(k)}$ (accuracy) by true class and rater across the four (α, β) configurations.

Thus, increasing J expands the parameter space substantially. Combined with sparse off-diagonal evidence in multi-class settings, this leads to weak identification and can manifest as multimodality and label switching, unless the model is sufficiently regularized.

The prior on class proportions is kept weak,

$$p \sim \text{Dirichlet}(\alpha \mathbf{1}), \quad \alpha = 1,$$

so that the data can determine the mixture weights. This choice is symmetric in the classes and therefore does not, by itself, resolve label switching. A comparison against alternative α specifications did not lead to noticeable differences in the three-category ISCO fits, hence only the $\alpha = 1$ results are shown.

The key regularization is the prior on each confusion-matrix row,

$$\pi_{j\cdot}^{(k)} \sim \text{Dirichlet}(\beta_{j\cdot}).$$

In this specification, off-diagonal components of $\beta_{j\cdot}$ are set to 1, and the diagonal component is set to β_{diag} . The induced prior mean of the diagonal entry is

$$\mathbb{E}[\pi_{jj}^{(k)}] = \frac{\beta_{\text{diag}}}{\beta_{\text{diag}} + (J - 1) \cdot 1}.$$

Holding the target prior mean near 0.83 implies that β_{diag} must increase with $J - 1$,

$$\beta_{\text{diag}} = \frac{0.83}{1 - 0.83}(J - 1).$$

For $J = 2$, this implies $\beta_{\text{diag}} \approx 4.88 \approx 5$. For $J = 3$, it implies $\beta_{\text{diag}} \approx 9.76 \approx 10$. In practice, $\beta_{\text{diag}} = 10$ still led to label switching in the three-category setting. Stable sampling without label switching was obtained once the diagonal mass was increased further, with stability reached from $\beta_{\text{diag}} = 15$ onward. To demonstrate effects and results, the three-category version is applied to ISCO data.

4.5.2 ISCO results (3 categories)

Only the informative configuration is reported, corresponding to $\alpha = \mathbf{1}$ and a diagonal-favoring confusion-matrix prior with $\beta_{\text{diag}} = 15$ and off-diagonals set to 1.

Figure 10 shows the posterior mean class proportions. In the three-category fit, class “2” dominates the prevalence while the remaining probability mass is split across the other two classes, indicating that the additional category is used but remains substantially less frequent.

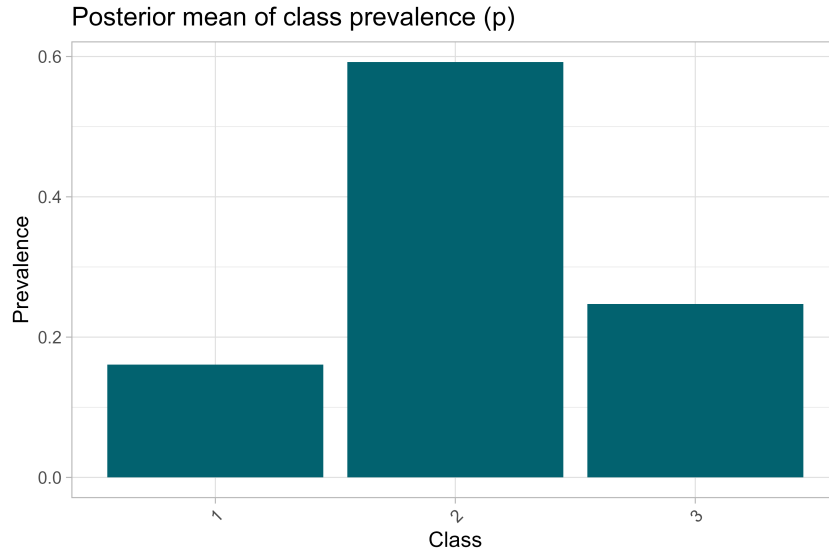


Figure 10: ISCO three-category fit (informative β , $\beta_{\text{diag}} = 15$). Posterior mean of class proportions p .

Rater performance can again be summarized compactly by the weighted diagonal mass $\sum_{j=1}^J p_j \pi_{jj}^{(k)}$. Figure 11 indicates that both human raters are close to diagonally dominant across the latent classes, while the tool shows a lower diagonal mass, consistent with higher misclassification probability in this three-category regime.

Table 11 reports the posterior mean confusion matrices. Both human raters are strongly diagonally dominant across all three latent classes, with diagonal entries consistently above 0.90. The automated tool shows substantially more off-diagonal mass in classes 1 and 3, while class 2 remains relatively accurate. This contrast is particularly visible for the cell True = 1 / Given = 1: the tool achieves 0.766 (76.6%), whereas both human raters exceed 0.90 (> 90%).

Table 11: ISCO three-category fit (informative β , $\beta_{\text{diag}} = 15$). Posterior mean confusion matrices $\Pi^{(k)}$ for both human raters and the tool. Rows correspond to the latent true class, columns to the observed label.

Rater	True	Given 1	Given 2	Given 3
Human 1	1	0.911	0.071	0.017
Human 1	2	0.013	0.942	0.045
Human 1	3	0.059	0.058	0.882
Human 2	1	0.908	0.074	0.018
Human 2	2	0.007	0.979	0.013
Human 2	3	0.022	0.018	0.960
Tool	1	0.766	0.141	0.093
Tool	2	0.037	0.919	0.044
Tool	3	0.068	0.114	0.818

4.5.3 Comparison: 2 vs 3 categories

Moving from $J = 2$ to $J = 3$ changes the inference problem in two connected ways.

First, the number of free parameters increases from $(J - 1)(1 + KJ)$, which equals 7 for $J = 2$ and $K = 3$, to 20 for $J = 3$ and $K = 3$. At fixed sample size, this reduces the information per parameter and makes weakly identified directions more likely.

Second, the confusion-matrix prior becomes materially more important. If off-diagonals of β_j are kept at 1 and one wants the same diagonal prior mean across different J , then β_{diag} must scale with $J - 1$. Even with this scaling, the three-category fits remained practically vulnerable to label switching at $\beta_{\text{diag}} = 10$. Stable behavior was achieved once the diagonal mass was increased further, with stability reached from $\beta_{\text{diag}} = 15$ onward. By contrast, changing α away from $\alpha = \mathbf{1}$ did not materially affect convergence or posterior summaries in this setting, so the reported results focus on the informative β configuration.

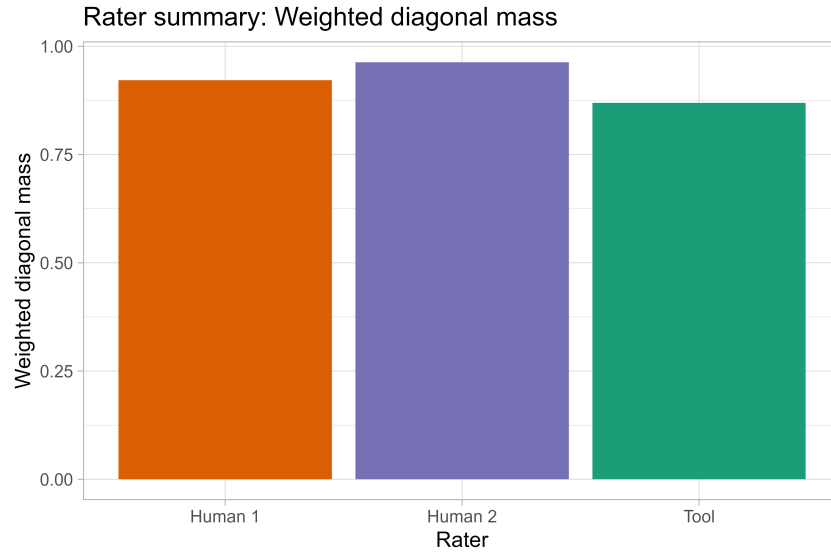


Figure 11: ISCO three-category fit (informative β , $\beta_{\text{diag}} = 15$). Rater summary based on the posterior mean confusion matrices: weighted diagonal mass $\sum_{j=1}^J p_j \pi_{jj}^{(k)}$.

5 Uncertainty quantification

5.1 Quantifying uncertainty at the item level

For each item i , the model provides posterior class probabilities

$$\omega_{i,j} = P(C_i = j \mid \text{data}), \quad j \in \{1, 2\},$$

which summarizes how plausible each latent class is given all observed ratings and the estimated rater error rates.

Item-level uncertainty is summarized using three complementary measures.

Maximum posterior probability.

$$m_i = \max_j \omega_{i,j}.$$

Values close to 1 indicate that one class dominates the posterior mass; values closer to 0.5 indicate borderline cases in the two-class setting.

Posterior entropy.

$$H_i = - \sum_{j=1}^2 \omega_{i,j} \log(\omega_{i,j}),$$

which equals 0 for deterministic assignments and is maximized at $\log(2)$ when $\omega_{i,1} = \omega_{i,2} = 0.5$.

Credible set size. For operational decisions, it is useful to summarize how many classes are needed to cover a fixed amount of posterior mass. The 95% credible set is the smallest set of classes whose posterior probabilities sum to at least 0.95. In the two-class case, its size is either 1, meaning one class already exceeds 0.95, or 2, meaning neither class reaches 0.95 and the item is genuinely ambiguous.

5.1.1 ISCO, two-category subset

Uncertainty quantification is first illustrated for ISCO restricted to major groups 2 and 3 under $\alpha = (1, 1)$, uniform β . While posterior distributions are highly concentrated for most items, a small set of borderline cases remains distinguishable. The credible set summary makes this explicit: 353 out of 357 items have a 95% credible set of size 1, while only 4 items require both classes to reach 95% posterior mass. These 4 cases are the natural candidates for targeted manual review.

Figure 12 shows the distribution of m_i . Most items lie close to 1, but the left tail contains a few uncertain cases. Figure 13 provides a complementary view via entropy, where values close to 0 indicate near-deterministic assignments.

To make the borderline structure tangible, Table 12 lists the most uncertain items ranked by entropy. A common characteristic of these cases is that the two human raters disagreed with each other, confirming that high posterior entropy successfully identifies genuinely ambiguous descriptions.

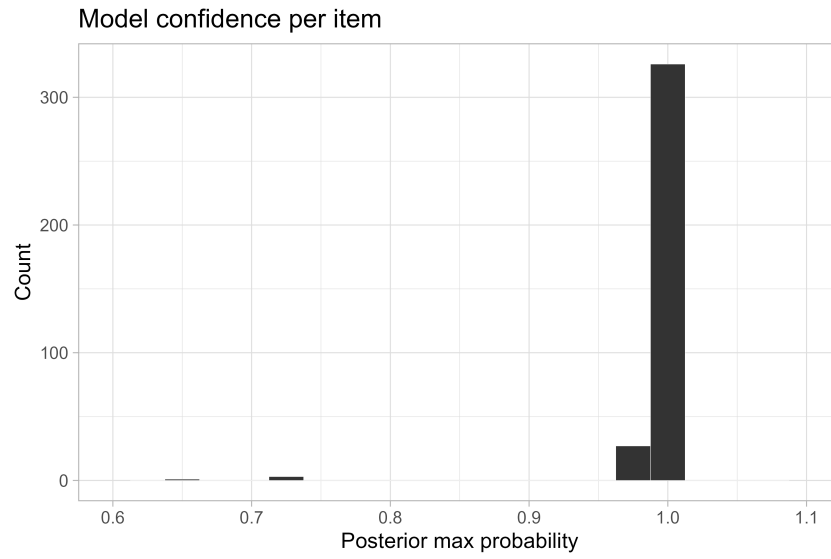


Figure 12: ISCO, two categories: distribution of per-item confidence $m_i = \max_j \omega_{i,j}$ for the baseline model ($\alpha = (1, 1)$, uniform β).

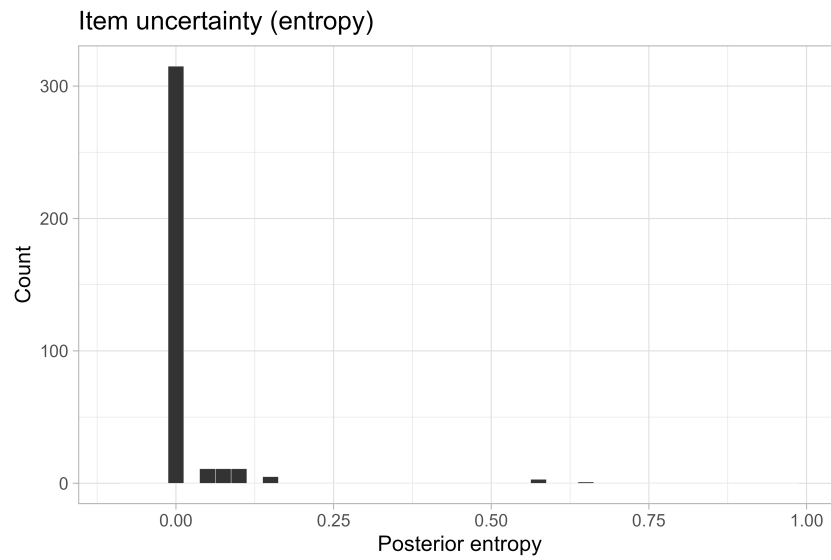


Figure 13: ISCO, two categories: distribution of posterior entropy H_i for the baseline model ($\alpha = (1, 1)$, uniform β).

Table 12: ISCO, most uncertain items by posterior entropy: baseline model ($\alpha = (1, 1)$, uniform β)

Item ID	$\hat{C}_i$	Posterior Statistics				Ratings		
		$\omega_{i,2}$	$\omega_{i,3}$	m_i	H_i	H1	H2	Tool
285	3	0.350	0.650	0.650	0.647	3	2	3
132	2	0.734	0.266	0.734	0.579	2	3	2
234	2	0.734	0.266	0.734	0.579	2	3	2
307	2	0.734	0.266	0.734	0.579	2	3	2

5.1.2 KldB, two-category subset

For KldB restricted to major groups 7 and 8, item-level uncertainty is essentially absent under the fitted specification. All 471 items have a 95% credible set of size 1, meaning one latent class already exceeds 0.95 posterior mass for every single item.

This is consistent with the distribution of m_i , which concentrates extremely closely to 1.

In practice, this means that for the KldB 7 versus 8 reduction, uncertainty-based triage provides little added value because there are virtually no borderline items at the item level.

5.2 Triage strategies and review prioritization

The posterior item distributions enable a simple review workflow. The motivation is to make practical use of uncertainty quantification: instead of treating all items equally, limited manual review capacity can be allocated to the cases where the model itself indicates ambiguity. A triage strategy is therefore proposed; that is, a rule-based procedure that separates items into (i) a low-risk group that can be accepted based on the posterior mode or a consensus label, and (ii) a small high-risk group that is prioritized for manual review. In applied settings, this helps reduce review workload while concentrating quality assurance on the items that are most likely to be misclassified.

5.2.1 Triage Strategy

Two equivalent decision rules are considered for the two-class setting to identify items requiring manual review. These rules are model-based but do not introduce new modeling assumptions, as they only post-process the posterior output.

Confidence threshold on m_i . Item i is flagged for review if $m_i < \tau$, where $\tau \in (0.5, 1)$ is a threshold that controls the workload. A higher value for τ results in more items being flagged.

Credible-set rule. Item i is flagged for review if the 95% credible set has a size larger than 1. In the two-class case, this is equivalent to $m_i < 0.95$.

Operationally, these rules are embedded into a two-step workflow:

Step 1 (Posterior Uncertainty). Flag item i for manual review if the 95% credible set has a size larger than 1. For $J = 2$, this corresponds to $m_i < 0.95$. Otherwise, the unique class in the 95% credible set is accepted.

Step 2 (Observed Disagreement). If the item passes Step 1 but the observed ratings are not unanimous, a disagreement screen is applied. Depending on review capacity, two options are available:

- Review all non-unanimous splits; or
- Prioritize by the outlier mechanism using estimated reliabilities, such as the confusion-matrix diagonal mass, as a risk proxy. Disagreements that contradict the most reliable source are reviewed. If the outlier is the least reliable source, the item may be accepted without review.

This strategy is attractive because it targets cases where the model cannot concentrate 95% posterior mass on a single class, ensuring that manual effort is focused where it is most needed. By choosing the credible-set level and the disagreement screen, the resulting review workload can be adjusted to available resources.

5.2.2 ISCO triage

For the ISCO dataset (Major Groups 2 versus 3), the 95% credible set rule yields a sharp separation. Only 4 items have a credible set of size 2, while the other 353 items have a unique class in the 95% credible set.

The results suggest that the model-based classification is sufficiently certain to bypass manual review for the vast majority of cases. If similar borderline cases recur, these findings can serve as feedback for the coding process, for example by refining decision rules or documenting clearer criteria for distinguishing the categories.

5.2.3 KldB triage

For KldB (7 versus 8), the credible set rule flags no items because all credible sets have a size of 1. Hence, a triage mechanism based on posterior uncertainty would rarely trigger and is not a meaningful lever in this two-category reduction. Instead, quality assurance should focus on aggregate rater behavior or on a more granular coding level if uncertainty-based review is desired.

6 Extensions and Discussion

6.1 Inclusion of covariates in the latent class model

The latent class model assumes a global prevalence vector p , that is, $P(C_i = j) = p_j$ for all items i . If item-specific covariates are available, for example, the respondent's education level, a natural extension is to let the latent class prevalence depend on these covariates. This is often referred to as latent class regression (Vermunt, 2010).

Covariate-dependent prevalence. Let x_i denote a vector of covariates for item i (including an intercept). In the two-class setting, the probability of one latent class is modeled via logistic regression.

$$P(C_i = 2 \mid x_i) = \text{logit}^{-1}(x_i^\top \beta), \quad P(C_i = 3 \mid x_i) = 1 - P(C_i = 2 \mid x_i).$$

Compared to the original model, the only change is that the global prevalence parameter p is replaced by the item-specific probabilities implied by x_i .

The observation model remains unchanged. Conditional on C_i , the rater model remains unchanged. For each rater k ,

$$P(Y_{i,k} = \ell \mid C_i = j) = \pi_{j,\ell}^{(k)},$$

so that covariates affect the posterior class probabilities through the prior odds of the latent class, while the rater confusion matrices still describe rater-specific error patterns.

Interpretation. The coefficient vector β quantifies how covariates shift the odds of belonging to one latent class versus the other. For example, a positive coefficient for education implies that higher education increases the probability of one major group, all else being equal. This extension is particularly useful for borderline items with rater disagreement, where covariates can stabilize posterior class assignments.

Practical considerations. Adding covariates increases model flexibility but also reduces the effective information per parameter. In settings with limited data or complex covariate structures, regularizing priors on β might be beneficial to prevent overfitting and ensure stable estimates.

ISCO is defined in terms of tasks and duties, so, normatively, coding should not be based on education. In practice, however, coders often use education as a heuristic because it correlates with many occupations and helps resolve ambiguous descriptions. This creates a tension for covariate extensions of the model: adding education as a predictor in the latent class model can improve predictive fit, but it may also encode a decision shortcut that is not part of the intended ISCO construct. Put differently, the model might learn $P(C_i \mid x_i)$ for a covariate x_i that is partly acting as a proxy for how coders behave rather than for the underlying occupational class itself. If the aim is to approximate an ISCO assignment that is faithful to the standard, covariates should ideally be restricted to information that is conceptually admissible, for example variables extracted from the

job description that directly reflect tasks, responsibilities, industry, or tools used, rather than background characteristics such as education.

From the measurement perspective, education can also affect rater behavior, not only the latent class distribution. In the basic noisy measurement model, each mechanism k has a single confusion matrix $\Pi^{(k)}$ with entries $\pi_{jl}^{(k)} = P(Y^{(k)} = l \mid C = j)$ that are assumed constant across items. If coders rely on education differently across education levels, then misclassification probabilities can become covariate dependent, meaning $P(Y^{(k)} = l \mid C = j, x) \neq P(Y^{(k)} = l \mid C = j)$. In that case, a constant $\Pi^{(k)}$ represents an average over heterogeneous error rates and can mask systematic patterns, for instance a tendency to upgrade certain classes when education is high. A principled extension would let error rates vary with covariates, such as modeling selected rows of $\Pi^{(k)}$ via a multinomial-logit regression on x , or at least performing sensitivity checks with and without education as a predictor. Either way, it is worth being explicit about the estimand: whether covariates are used to better approximate an ISCO-consistent latent class assignment, or to model and potentially diagnose rater heuristics that may depart from the normative coding rules.

6.2 Relaxing Dawid and Skene model assumptions

The classical Dawid and Skene model relies on two core independence assumptions, namely the conditional independence of ratings given the true class and the absence of item by observer interaction (Dawid and Skene, 1979). In addition, it assumes a single latent true class per item and a global prevalence vector that is shared across items. These assumptions yield a tractable model and are often a reasonable first approximation, but it is worth considering how they could be relaxed when substantive knowledge or richer data suggest systematic deviations. In this section, four natural directions of relaxation are the focus, and it is briefly discussed why they can be practically challenging in occupational coding settings.

(1) Single true class per item. The Dawid and Skene model assumes that each item belongs to exactly one latent class. Ambiguity is consequently represented only through posterior uncertainty over discrete categories. A conceptually appealing relaxation is to allow items to be intrinsically ambiguous, meaning that even under perfect measurement, the same case could plausibly map to more than one category. In practice, this can be addressed by introducing explicit item ambiguity parameters or by utilizing models in which item properties, such as difficulty, explain systematic borderline behavior (Paun et al., 2018). The practical hurdles to this approach are interpretability and identifiability. Once items are permitted to be partly multiclass, it becomes more difficult to distinguish intrinsic ambiguity from rater error without additional information. Such information might include adjudicated labels, richer item text features, or a high number of repeated ratings per item.

(2) Global prevalence is shared across items. In the original model, class proportions are constant across items, which is often unrealistic when covariates that are strongly related to occupational group membership, for example education, are available. A standard relaxation would be achieved by the latent class regression discussed in chapter

6.1, where class membership depends on covariates via a (multi)nomial logistic regression model, while the measurement model remains unchanged (Vermunt, 2010). This extension is attractive because it is minimal in spirit; it replaces only the constant mixture weights with item specific probabilities. The main practical hurdles are data sparsity and model building overhead: as the number and complexity of covariates increase, estimation can become unstable (e.g., due to sparse cells or (quasi-)separation), which typically motivates the use of regularizing priors and careful sensitivity checks (Gelman et al., 2008).

(3) Conditional independence of ratings given the true class. Dawid and Skene explicitly assume that, given the true response, ratings are independent across observers and repeated measurements (Dawid and Skene, 1979). Formally, for two mechanisms $k \neq k'$ this assumes $Y_i^{(k)} \perp Y_i^{(k')} \mid C_i$. In applied annotation tasks, this can fail; for example, when raters share training materials, follow the same tool suggestions, or influence each other. A common relaxation in latent class analysis is to allow *local dependence* between selected pairs of indicators (Vermunt, 2010). In the present setting, the indicators are the ratings themselves (tool and human coders), and local dependence means that for some pairs (k, k') the joint distribution does not factorize given C_i , i.e., $P(Y_i^{(k)}, Y_i^{(k')} \mid C_i) \neq P(Y_i^{(k)} \mid C_i) P(Y_i^{(k')} \mid C_i)$. Such local dependencies can bias inference if ignored and typically require explicit modeling or careful diagnostic checks (Berzofsky et al., 2014). The practical hurdle is that adding dependence parameters increases model complexity and can lead to estimation and convergence problems in applied settings (Nylund-Gibson and Choi, 2018), while latent class models can also exhibit non-identifiability and difficult likelihood geometry even in seemingly simple cases (Fienberg et al., 2007).

(4) No item by observer interaction; constant error rates per rater. Dawid and Skene also assume that there is no patient-clinician interaction, which corresponds to constant rater error profiles that do not vary by item (Dawid and Skene, 1979). A widely used relaxation is to introduce item difficulty effects, such that the probability of a correct label depends on both a rater ability component and an item difficulty component. This aligns with the broader family of item response theory models and is explicitly discussed in modern comparisons of Bayesian annotation models, where item difficulty is one of the key extensions beyond Dawid and Skene (Paun et al., 2018). The practical hurdle is parameter growth: adding one difficulty parameter per item can dominate the information in small to medium data sets, and it can become difficult to disentangle true item difficulty from systematic rater bias without strong priors or substantially more repeated ratings.

Relaxations of Dawid and Skene assumptions are well motivated and, in parts, widely studied, especially regarding covariate dependent prevalence and item difficulty (Dawid and Skene, 1979, Vermunt, 2010, Paun et al., 2018). However, they typically require more data overlap, more parameters, and more careful regularization than the original model. For the present application, these extensions are best viewed as conceptual directions for future work, to be prioritized when moving to more granular coding levels or when richer item covariates and higher redundancy of ratings are available.

6.3 Beyond Occupational Coding: Origins and Conceptual Horizons

The principle of modeling rather than ignoring disagreement through overlapping labels, which served as the foundation for the occupational coding analysis, extends naturally to a variety of other high-impact domains. In these settings, as in occupational classification, quantifying uncertainty allows for an informed aggregation of sources and a targeted allocation of review effort. This approach is especially valuable where labeling decisions are costly or carry significant consequences.

The Dawid and Skene framework was originally introduced in a medical measurement context and illustrated with a pre-operative assessment task: five anesthetists independently classified 45 patients on a four-point “fitness for anesthesia” scale, with one anesthetist repeating the assessment three times (Dawid and Skene, 1979). In this setting, the model yields a probabilistic consensus for each patient and estimates clinician-specific error profiles that can be used for monitoring and quality assurance (Dawid and Skene, 1979). A natural operational use could be triage: based on the posterior over the latent class, cases with concentrated posteriors can be handled routinely, while cases with diffuse posteriors are prioritized for additional diagnostics, second opinions, or adjudication. Importantly, such a triage policy does not necessarily require re-estimating the model after every new rating. In practice, one would fit the model on a recent batch of overlapping labels, compute posterior uncertainties for incoming cases, route borderline cases to expert review, and periodically refit or update the model once enough new adjudicated labels or reviewer decisions have accumulated.

The same logic of uncertainty-driven triage underpins the semi-automated coding of open-ended survey questions. In this field, manual coding is expensive and reliability is typically assessed only on a small double-coded subset. Schonlau and Couper (2016) propose a semi-automatic workflow in which a statistical classifier codes easy cases and humans code the remainder, using a threshold to control the manual workload. He and Schonlau (2022) describe a complementary triage goal: when additional double-coding is possible, one can prioritize cases with high estimated risk of coding error (He and Schonlau, 2022). Both settings match the logic of noisy measurement models: disagreement is expected, uncertainty can be quantified case by case, and review capacity can be targeted at items where the evidence is least decisive.

A conceptual proposal for future research involves the potential application of latent class models to content moderation and annotation pipelines for large language models (LLMs). In such pipelines, labels often come from a mixture of crowdworkers, trained contractors, domain experts, and automated pre-classifiers, and disagreement is common because categories are context dependent and borderline cases are frequent. For example, OpenAI describes safety pipelines that combine automated classifiers and human review for categorization and enforcement decisions in the deployment of GPT-4 class models (OpenAI, 2023). Similarly, Meta describes moderation workflows in which potential violations are routed into reviewer queues for human evaluation, with technology and review teams jointly supporting enforcement at scale (Meta, 2024). In this setting, latent class models would provide (i) probabilistic aggregated labels, (ii) source specific error patterns,

and (iii) item-level uncertainty that can drive escalation policies: ambiguous items can be routed to higher expertise review or adjudication, while high-confidence items can be processed automatically.

7 Conclusion

This thesis studied occupational coding as a noisy label problem in which multiple imperfect sources provide categorical classifications without a gold standard. Building on the Dawid and Skene (1979) model, the analysis implemented a Bayesian noisy measurement formulation to represent latent true classes, class prevalence, and rater-specific confusion matrices. It explored how these quantities can be estimated and diagnosed within a probabilistic programming workflow using Stan, while acknowledging that the implementation serves as an application of existing frameworks rather than a new theoretical development.

Empirically, the model yielded interpretable insights for the classification schemes considered, though these findings are closely tied to the specific data context of the infas usability study. In the ISCO two-category subset (major groups 2 vs 3), posterior inference indicated a notable prevalence imbalance toward class 2 and suggested discernible differences in rater quality. Specifically, human rater 2 appeared to be in close alignment with the latent class, while the tool showed a directional weakness by disproportionately mapping true class 3 cases to observed class 2. Such patterns are of operational interest as they potentially affect the detection of minority occupational groups.

In the KldB two-category reduction (major groups 7 vs 8), the model suggested a more balanced prevalence structure. Both human raters appeared almost perfectly aligned with the latent classification, while the tool showed only mild asymmetry. In this setting, the limited disagreement suggests that posterior uncertainty provides less additional leverage for prioritization, as the observed structure remains near-deterministic.

A central contribution of this Bayesian approach is making uncertainty explicit. By using posterior item probabilities, the thesis outlined uncertainty measures that can be translated into triage rules for manual review. The application of the 95% credible-set rule to the ISCO data partly suggested a separation of borderline cases. However, the practical utility of such uncertainty-based triage appears to depend heavily on the intrinsic ambiguity of the specific coding task and the resulting disagreement structure.

Combinations of priors for the class proportions and confusion matrices were evaluated in a focused sensitivity grid. The comparison shows that posterior summaries can shift in nontrivial ways under stronger prevalence priors, so any apparent robustness within a narrow range should not be overstated. More fundamentally, all results inherit the core Dawid and Skene assumptions, in particular conditional independence of ratings given the latent class. If this is violated, for example through shared training, common guidance materials, tool-induced anchoring, or systematic item-level difficulty, then both inferred rater qualities and latent class assignments can be biased. Consequently, the reported metrics should be interpreted as model-based indications under these assumptions, not as absolute measures of rater accuracy.

Several limitations suggest that these findings represent an initial step. Moving beyond coarse two-category reductions to granular coding levels will likely require stronger pooling or structured priors to handle increased dimensionality.

Overall, the thesis suggests that Bayesian noisy measurement models offer a structured alternative to simple agreement summaries. Rather than replacing existing workflows, they provide a systematic basis for expert discussion, which remains the essential next step when coders disagree in occupational coding. By quantifying uncertainty and separating source bias from item ambiguity, these models can help inform quality assurance workflows and rater monitoring in a more nuanced manner.

A Appendix

A.1 Closed-form maximizers of the complete-data log-likelihood

This appendix derives the closed-form maximizers of the complete-data log-likelihood ℓ_c under the simplex constraints for p and the rows of each confusion matrix $\Pi^{(k)}$.

Setup. Let $T_{ij} \in \{0, 1\}$ indicate the latent true class of item i , with $\sum_{j=1}^J T_{ij} = 1$. Let $n_{il}^{(k)}$ denote how often rater k assigns label l to item i (in our data this is typically 0 or 1). The complete-data log-likelihood can be written as

$$\ell_c(p, \Pi) = \sum_{i=1}^I \sum_{j=1}^J T_{ij} \log p_j + \sum_{k=1}^K \sum_{j=1}^J \sum_{l=1}^J \left(\sum_{i=1}^I T_{ij} n_{il}^{(k)} \right) \log \pi_{jl}^{(k)}.$$

The constraints are $\sum_{j=1}^J p_j = 1$ and, for each j, k , $\sum_{l=1}^J \pi_{jl}^{(k)} = 1$.

Maximization w.r.t. p . Consider the Lagrangian

$$\mathcal{L}(p, \lambda) = \sum_{i=1}^I \sum_{j=1}^J T_{ij} \log p_j + \lambda \left(\sum_{j=1}^J p_j - 1 \right).$$

For each j ,

$$\frac{\partial \mathcal{L}}{\partial p_j} = \frac{\sum_{i=1}^I T_{ij}}{p_j} + \lambda = 0 \quad \Rightarrow \quad p_j = -\frac{1}{\lambda} \sum_{i=1}^I T_{ij}.$$

Imposing $\sum_{j=1}^J p_j = 1$ yields

$$1 = \sum_{j=1}^J p_j = -\frac{1}{\lambda} \sum_{j=1}^J \sum_{i=1}^I T_{ij} = -\frac{1}{\lambda} \sum_{i=1}^I \sum_{j=1}^J T_{ij} = -\frac{1}{\lambda} I,$$

hence $\lambda = -I$ and therefore

$$\hat{p}_j = \frac{1}{I} \sum_{i=1}^I T_{ij}.$$

Maximization w.r.t. $\pi_{jl}^{(k)}$. Fix j and k and maximize the terms involving the row $(\pi_{j1}^{(k)}, \dots, \pi_{jJ}^{(k)})$ subject to $\sum_{l=1}^J \pi_{jl}^{(k)} = 1$:

$$\mathcal{L}_{jk}(\pi_{j\cdot}^{(k)}, \mu_{jk}) = \sum_{l=1}^J \left(\sum_{i=1}^I T_{ij} n_{il}^{(k)} \right) \log \pi_{jl}^{(k)} + \mu_{jk} \left(\sum_{l=1}^J \pi_{jl}^{(k)} - 1 \right).$$

For each l ,

$$\frac{\partial \mathcal{L}_{jk}}{\partial \pi_{jl}^{(k)}} = \frac{\sum_{i=1}^I T_{ij} n_{il}^{(k)}}{\pi_{jl}^{(k)}} + \mu_{jk} = 0 \quad \Rightarrow \quad \pi_{jl}^{(k)} = -\frac{1}{\mu_{jk}} \sum_{i=1}^I T_{ij} n_{il}^{(k)}.$$

Summing over l and using the constraint gives

$$1 = \sum_{l=1}^J \pi_{jl}^{(k)} = -\frac{1}{\mu_{jk}} \sum_{l=1}^J \sum_{i=1}^I T_{ij} n_{il}^{(k)} \quad \Rightarrow \quad \mu_{jk} = -\sum_{l=1}^J \sum_{i=1}^I T_{ij} n_{il}^{(k)}.$$

Plugging back yields the closed-form estimator

$$\hat{\pi}_{jl}^{(k)} = \frac{\sum_{i=1}^I T_{ij} n_{il}^{(k)}}{\sum_{l'=1}^J \sum_{i=1}^I T_{ij} n_{il'}^{(k)}}.$$

Remark (connection to EM). The above maximizers are conditional on the latent indicators T_{ij} . In the EM algorithm, the M-step uses the same closed-form expressions after replacing T_{ij} by its posterior expectation $w_{ij} = \mathbb{E}[T_{ij} \mid \text{data}]$ computed in the E-step.

A.2 Occupational classification codes (1-digit levels)

Table 13: ISCO-08 major groups (1-digit, Major Groups)

ISCO-08 (1-digit)	Title
0	Armed Forces Occupations
1	Managers
2	Professionals
3	Technicians and Associate Professionals
4	Clerical Support Workers
5	Service and Sales Workers
6	Skilled Agricultural, Forestry and Fishery Workers
7	Craft and Related Trades Workers
8	Plant and Machine Operators and Assemblers
9	Elementary Occupations

Table 14: KldB 2010 major groups (1-digit, Berufsbereiche)

KldB 2010 (1-digit)	Description
0	Military
1	Agriculture, forestry, animal husbandry and horticulture
2	Raw material extraction, production and manufacturing
3	Construction, architecture, surveying and building services engineering
4	Natural sciences, geography and computer science
5	Transport, logistics, protection and security
6	Commercial services, trade, sales, hotel and tourism
7	Business organisation, accounting, law and administration
8	Health, social services, teaching and education
9	Languages, literature, humanities, social and economic sciences, media, art, culture and design

A.3 Example convergence diagnostics

A.3.1 Split- $\hat{R}$ overview and sample size

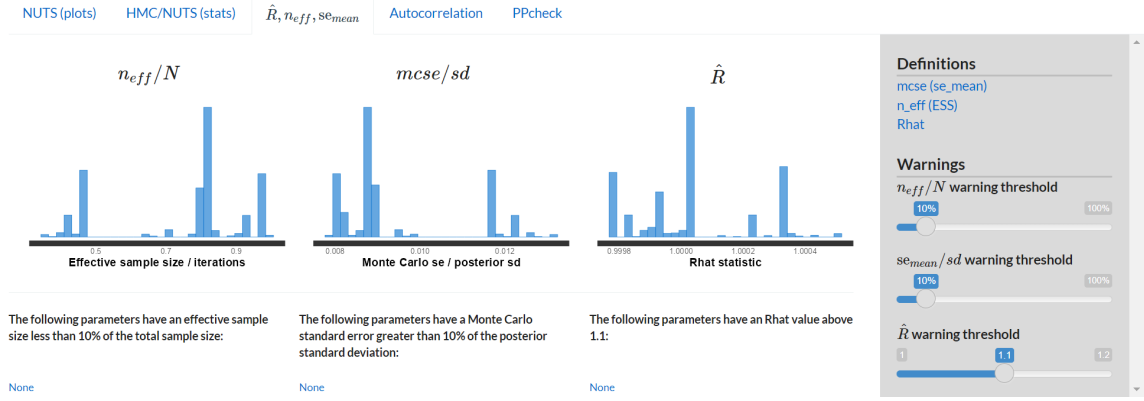


Figure 14: Split- $\hat{R}$ overview for the ISCO two-category model with non-informative priors ($\alpha = (1, 1)$, uniform β). Values concentrate very near 1, with no systematic inflation across parameter groups, indicating agreement across chains. Additionally, all parameters maintain an effective sample size that exceeds 10% of the total sample size.

A.3.2 Sampler diagnostics: Treedepth and Divergences

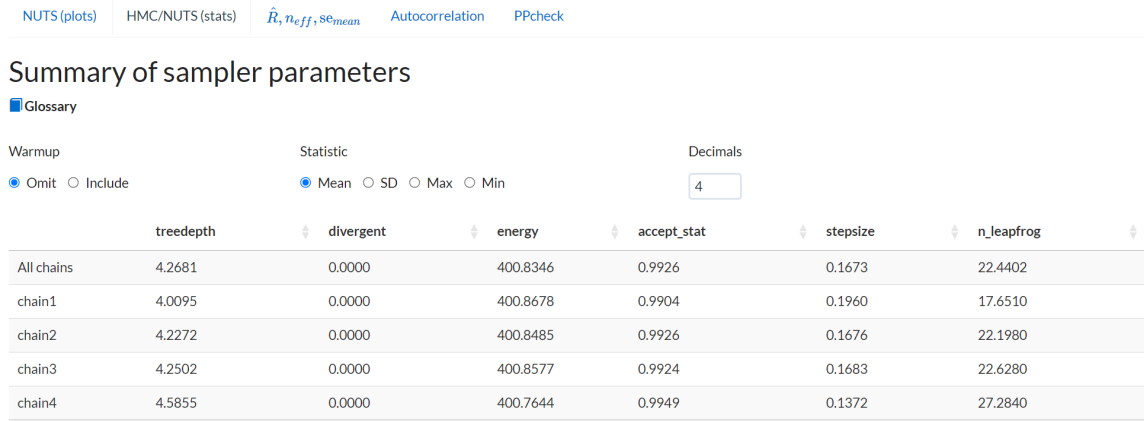


Figure 15: Sampler diagnostics for the ISCO two-category model with non-informative priors ($\alpha = (1, 1)$, uniform β). No divergent transitions are reported and treedepth remains well below the maximum, which indicates stable Hamiltonian trajectories for this fit.

A.3.3 Autocorrelation

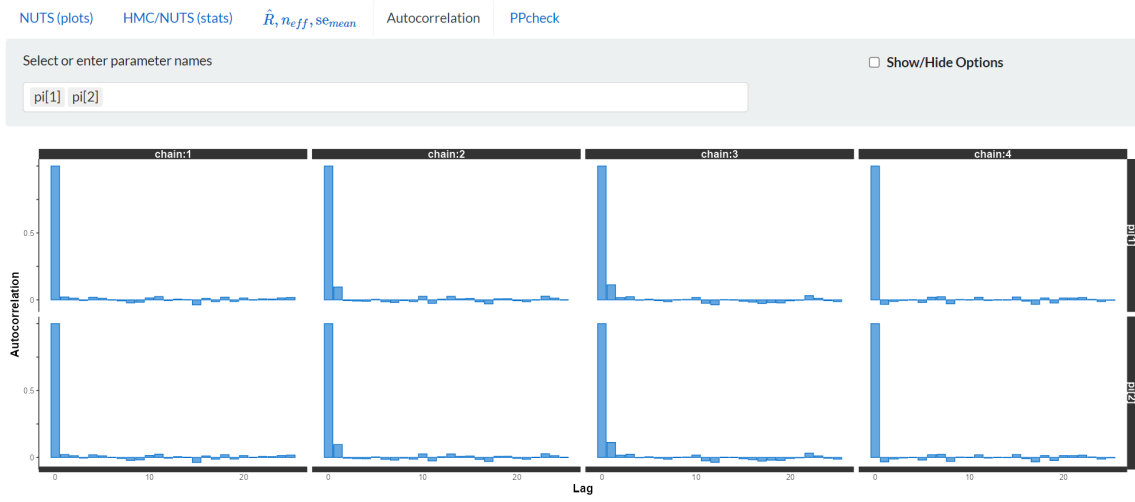


Figure 16: Autocorrelation diagnostics for the ISCO two-category model with non-informative priors ($\alpha = (1, 1)$, uniform β). Autocorrelations drop quickly for the displayed parameters, consistent with efficient mixing and adequate effective sample sizes.

A.4 Quick reference: Distributions used in this thesis

Throughout this thesis, discrete class variables and probability vectors are used. A probability vector over J classes lies on the simplex

$$\Delta^{J-1} = \left\{ p \in \mathbb{R}^J : p_j \geq 0, \sum_{j=1}^J p_j = 1 \right\}.$$

Categorical distribution The categorical distribution describes a single draw from J categories with probabilities $p \in \Delta^{J-1}$:

$$Y \mid p \sim \text{Categorical}(p), \quad P(Y = l \mid p) = p_l, \quad l = 1, \dots, J.$$

Multinomial distribution The multinomial distribution describes class counts obtained from N independent categorical draws with the same parameter p :

$$n = (n_1, \dots, n_J) \mid (N, p) \sim \text{Multinomial}(N, p), \quad \sum_{l=1}^J n_l = N,$$

$$P(n \mid N, p) = \frac{N!}{\prod_{l=1}^J n_l!} \prod_{l=1}^J p_l^{n_l}.$$

In the rater setting, $n_{il}^{(k)}$ can denote the number of ratings by rater k for item i in class l .

Dirichlet distribution The Dirichlet distribution is a distribution on a probability vector $p \in \Delta^{J-1}$ with concentration parameters $\alpha = (\alpha_1, \dots, \alpha_J)$, $\alpha_j > 0$:

$$p \sim \text{Dirichlet}(\alpha), \quad f(p) \propto \prod_{j=1}^J p_j^{\alpha_j-1}.$$

It is conjugate to categorical and multinomial models. In this thesis, Dirichlet priors are used, for example, for class prevalences p and for rows of misclassification matrices $\pi_{j\cdot}^{(k)}$, where $\pi_{j\cdot}^{(k)} \in \Delta^{J-1}$.

Beta distribution The beta distribution is a distribution on a probability $q \in (0, 1)$:

$$q \sim \text{Beta}(a, b), \quad f(q) \propto q^{a-1} (1-q)^{b-1}.$$

It is conjugate to the binomial distribution and is a special case of the Dirichlet distribution for $J = 2$.

B Electronic appendix

B.1 Codebase

The complete codebase, submission files, and extended figures are hosted in a dedicated GitLab repository:

`https://gitlab.lrz.de/000000000135BF5A/noisy\_measurement`

Please refer to the README file for an overview of the repository structure and usage instructions:

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Munich, January 27th, 2026

Julian Ehrenberg